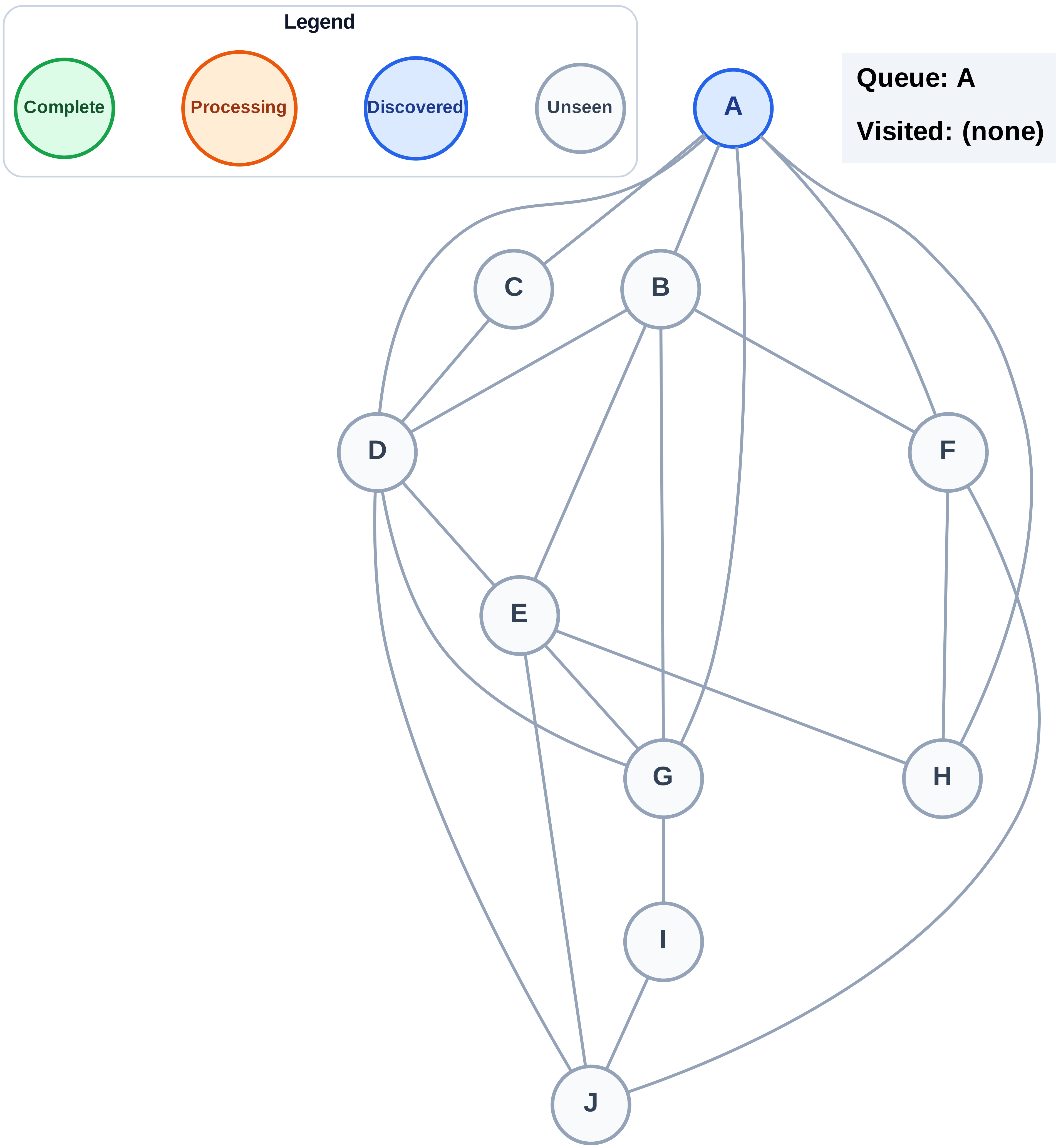
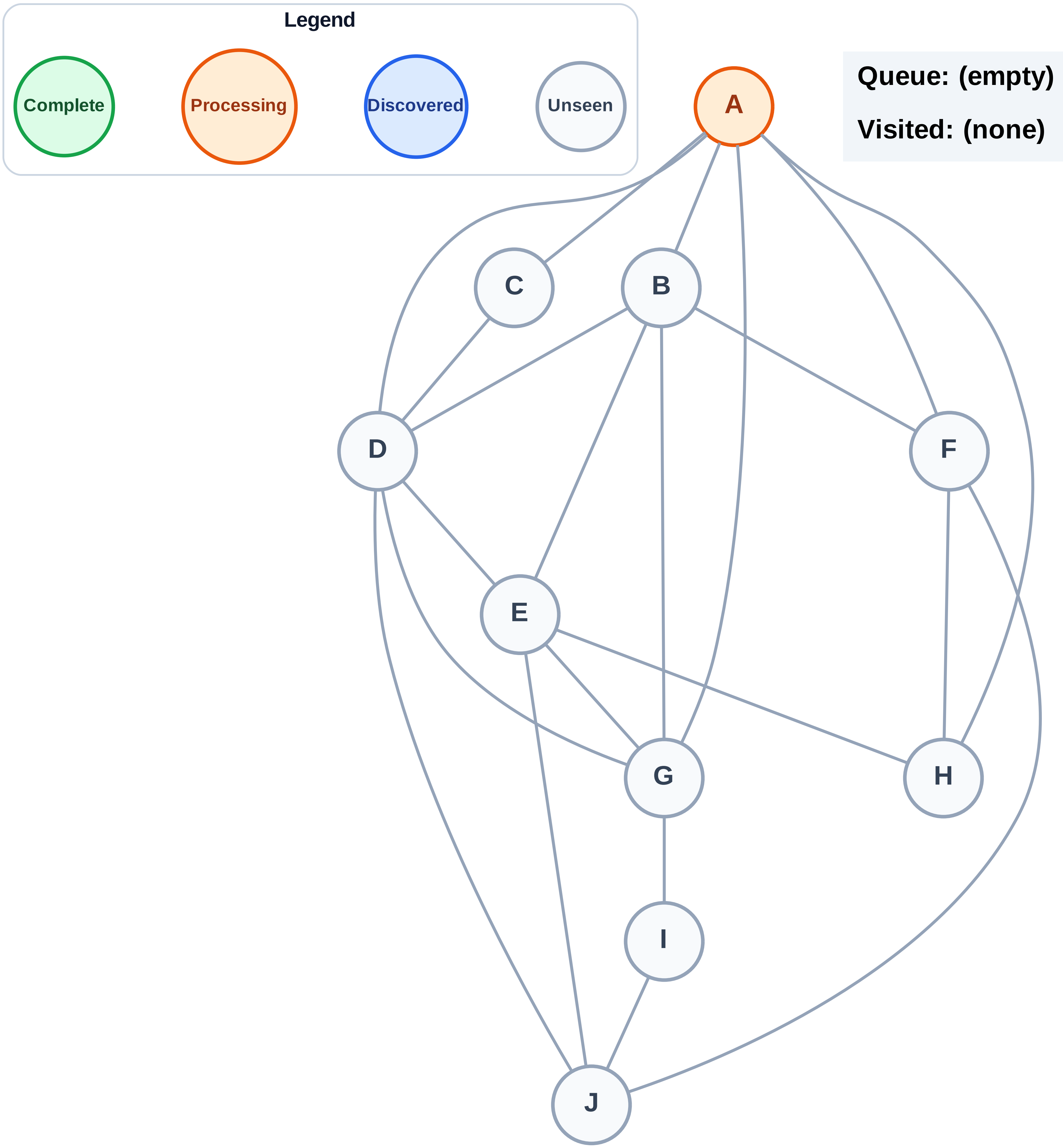
BFS Traversal

Step 1: Start at A



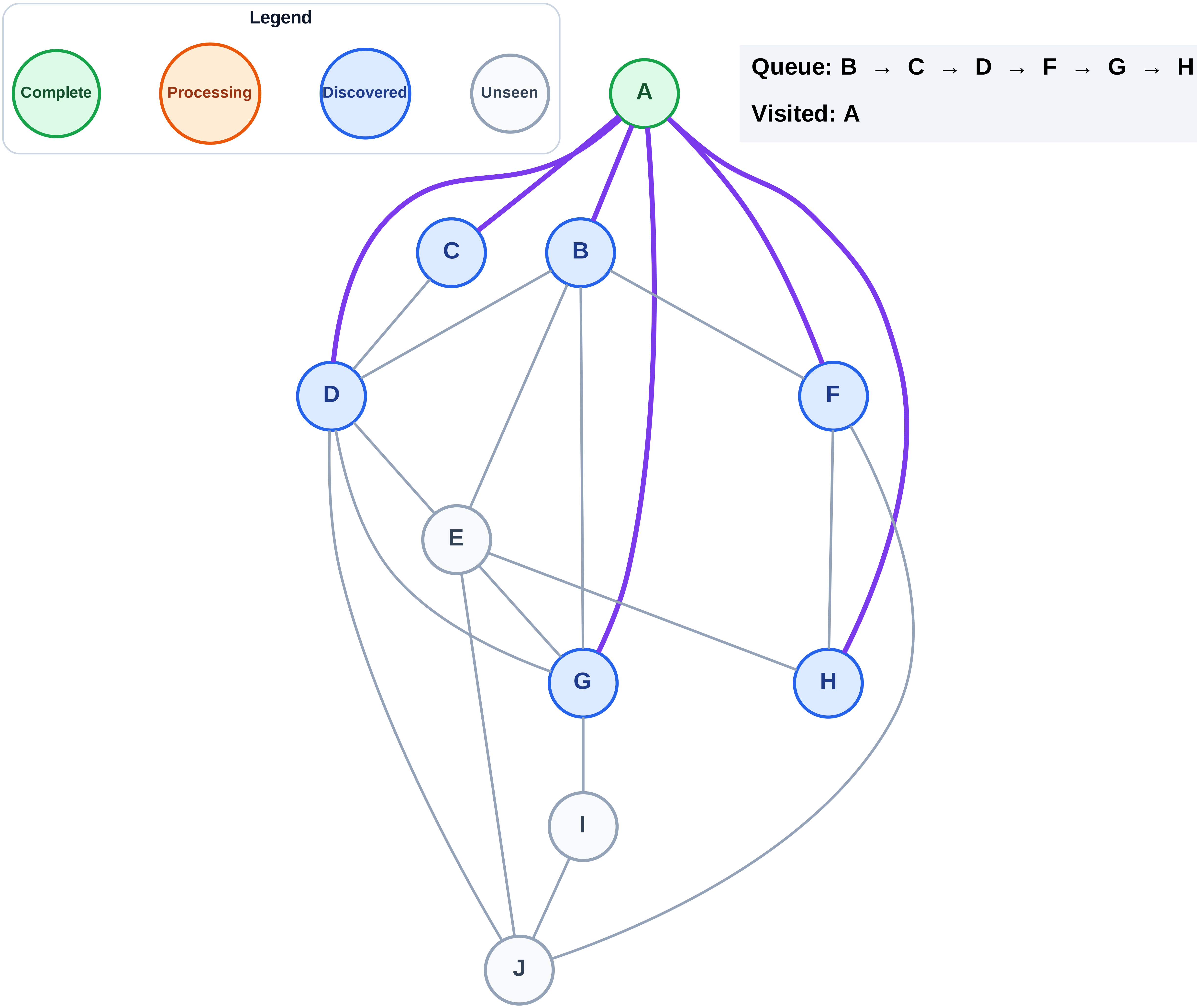
BFS Traversal

Step 2: Process node A



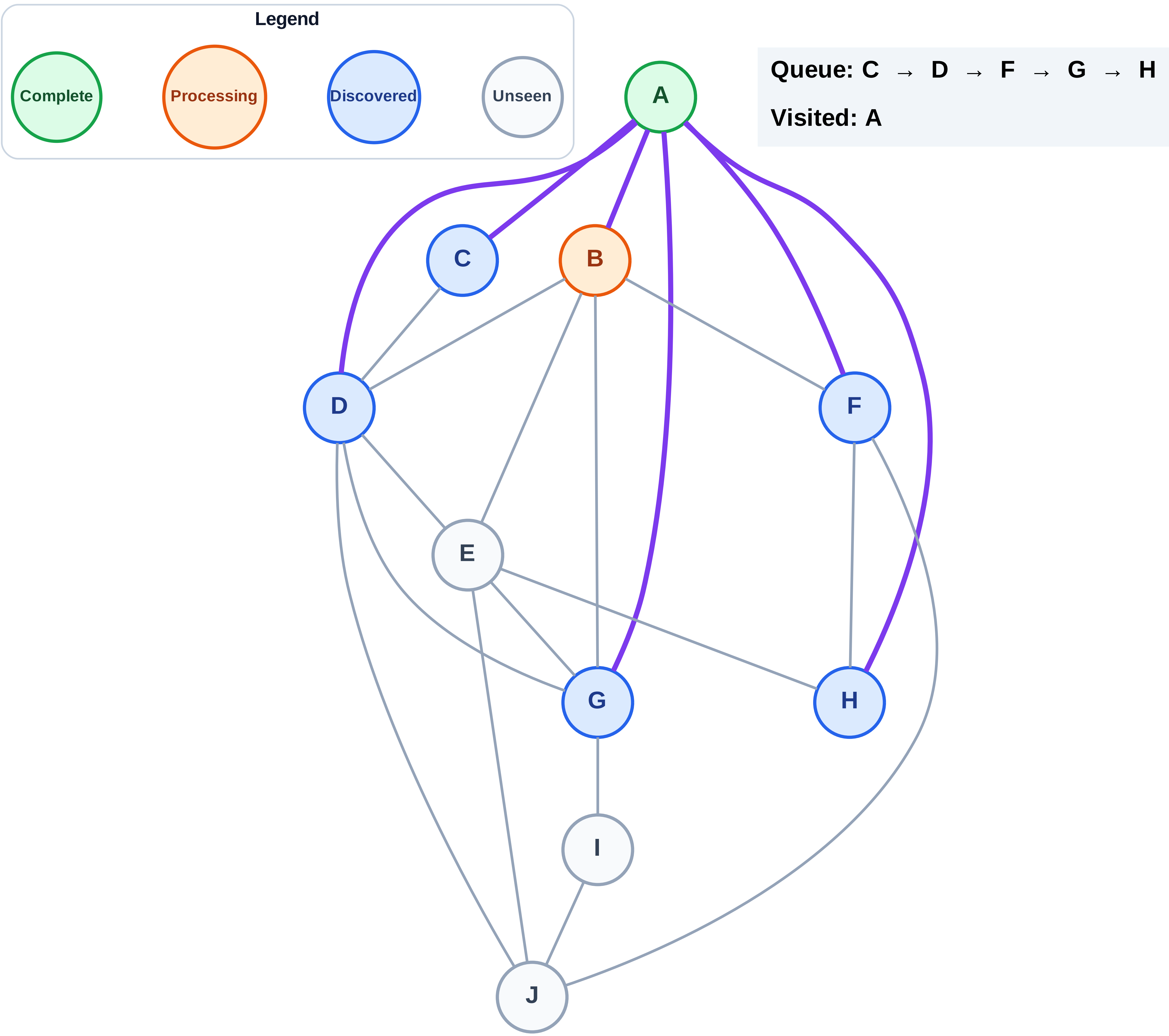
BFS Traversal

Step 3: Discover B, C, D, F, G, H from A



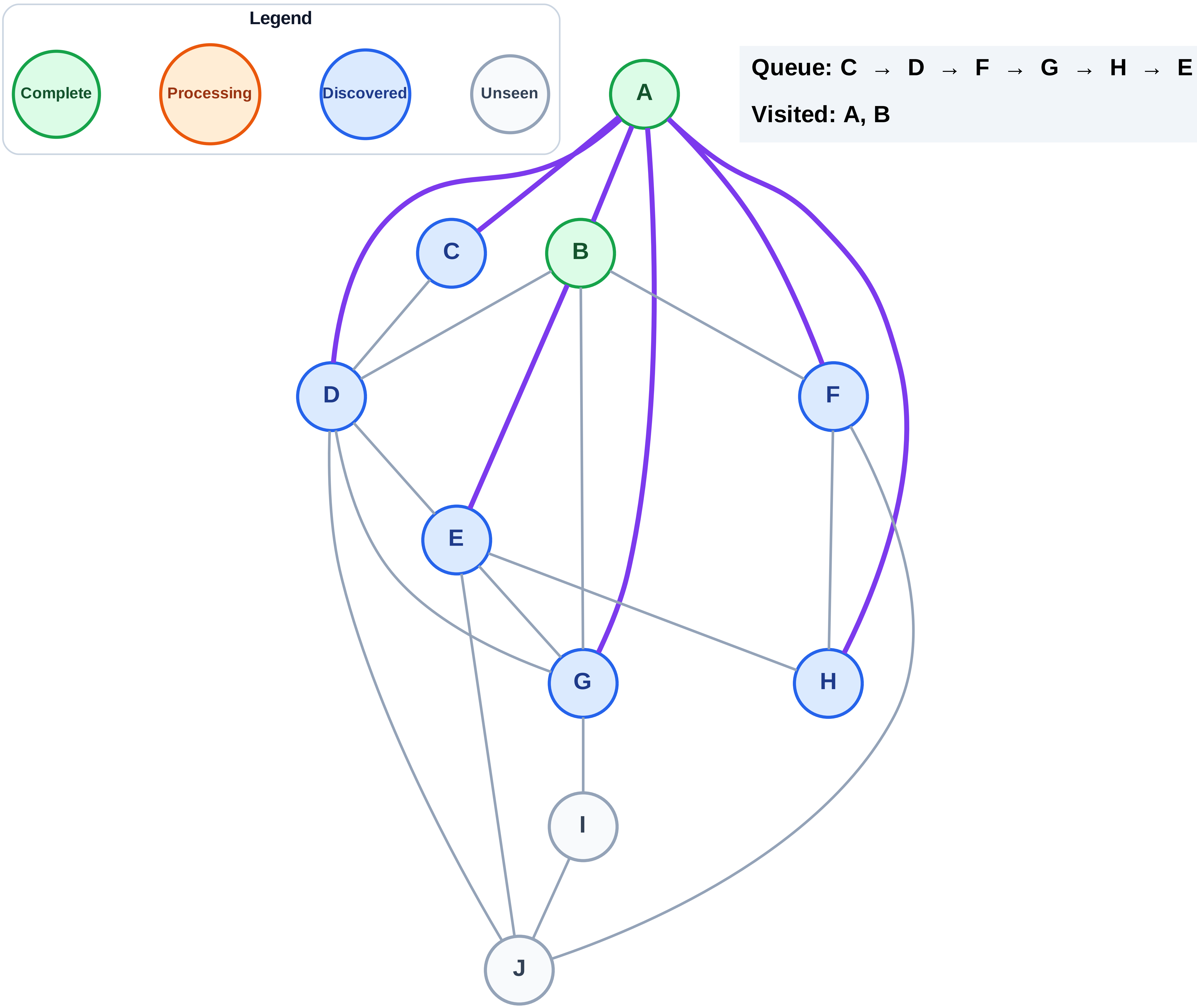
BFS Traversal

Step 4: Process node B



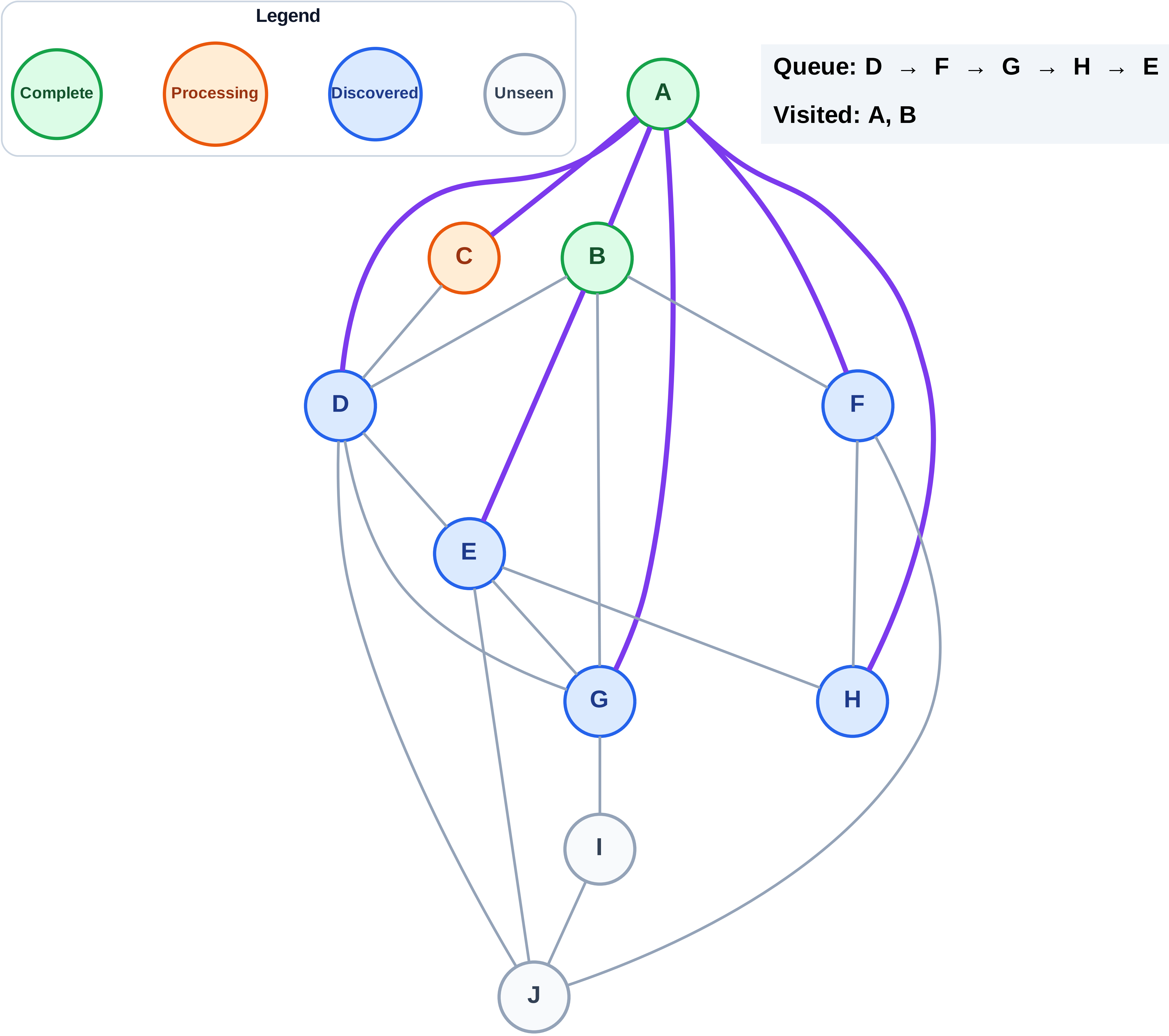
BFS Traversal

Step 5: Discover E from B



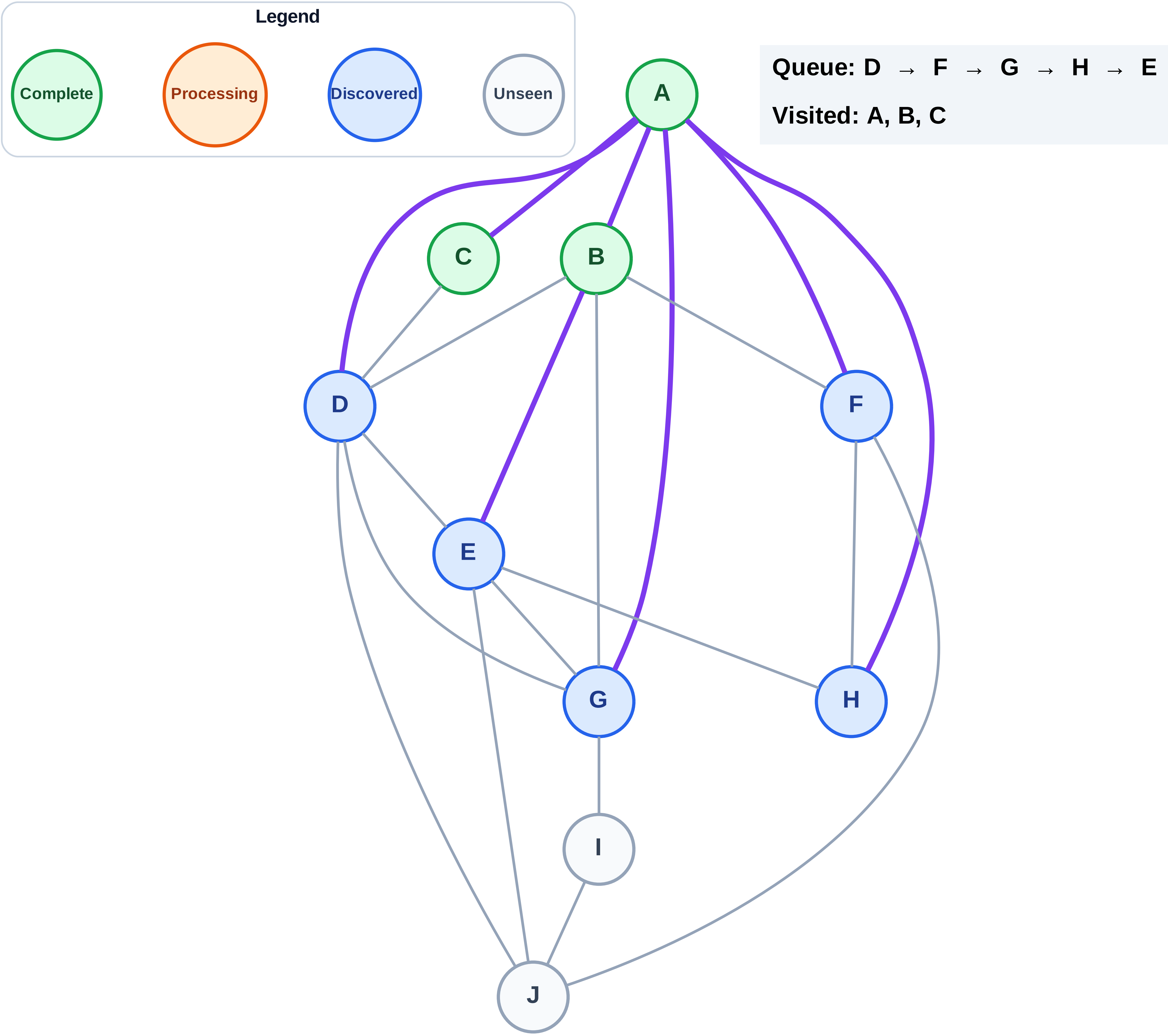
BFS Traversal

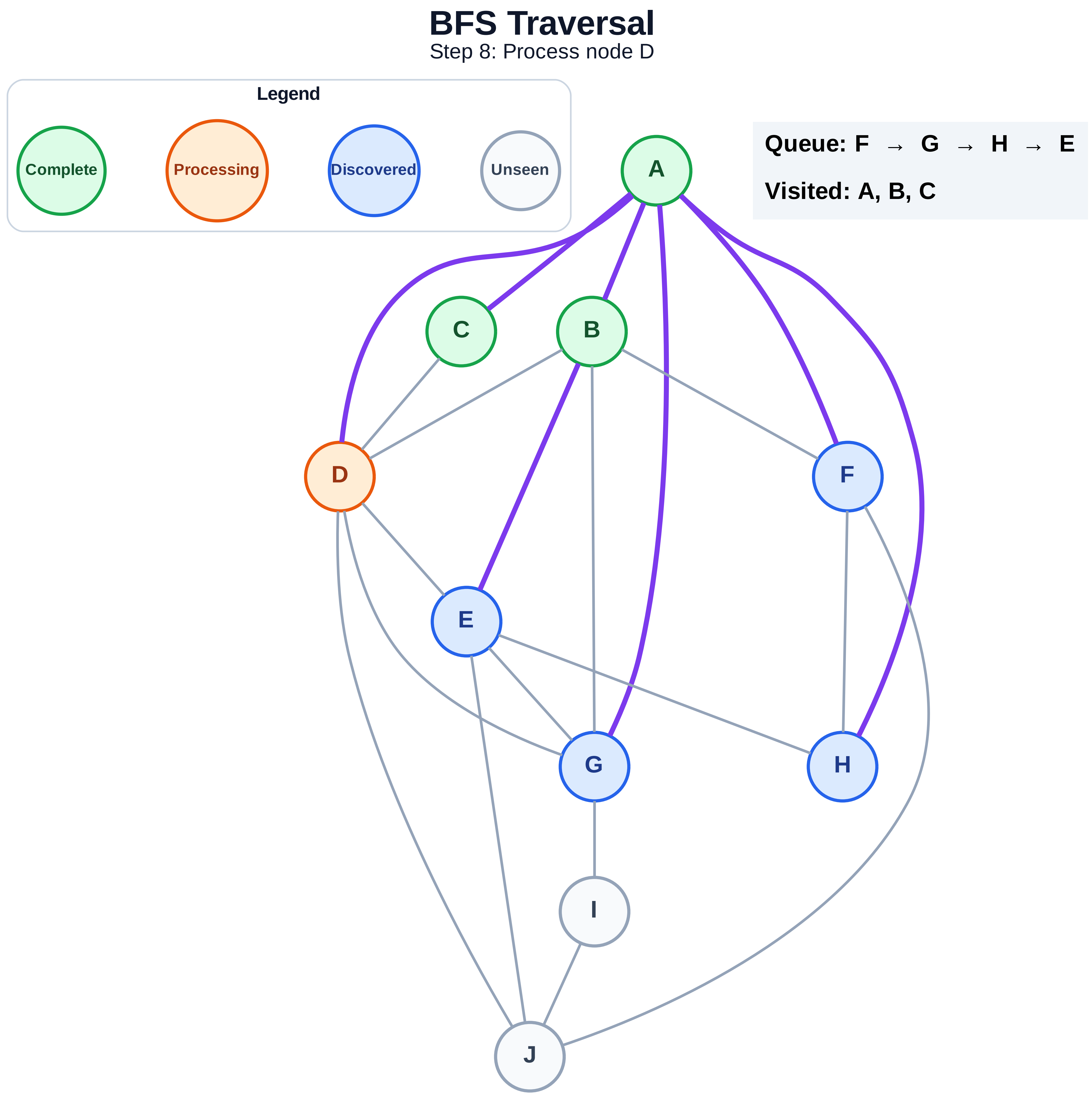
Step 6: Process node C



BFS Traversal

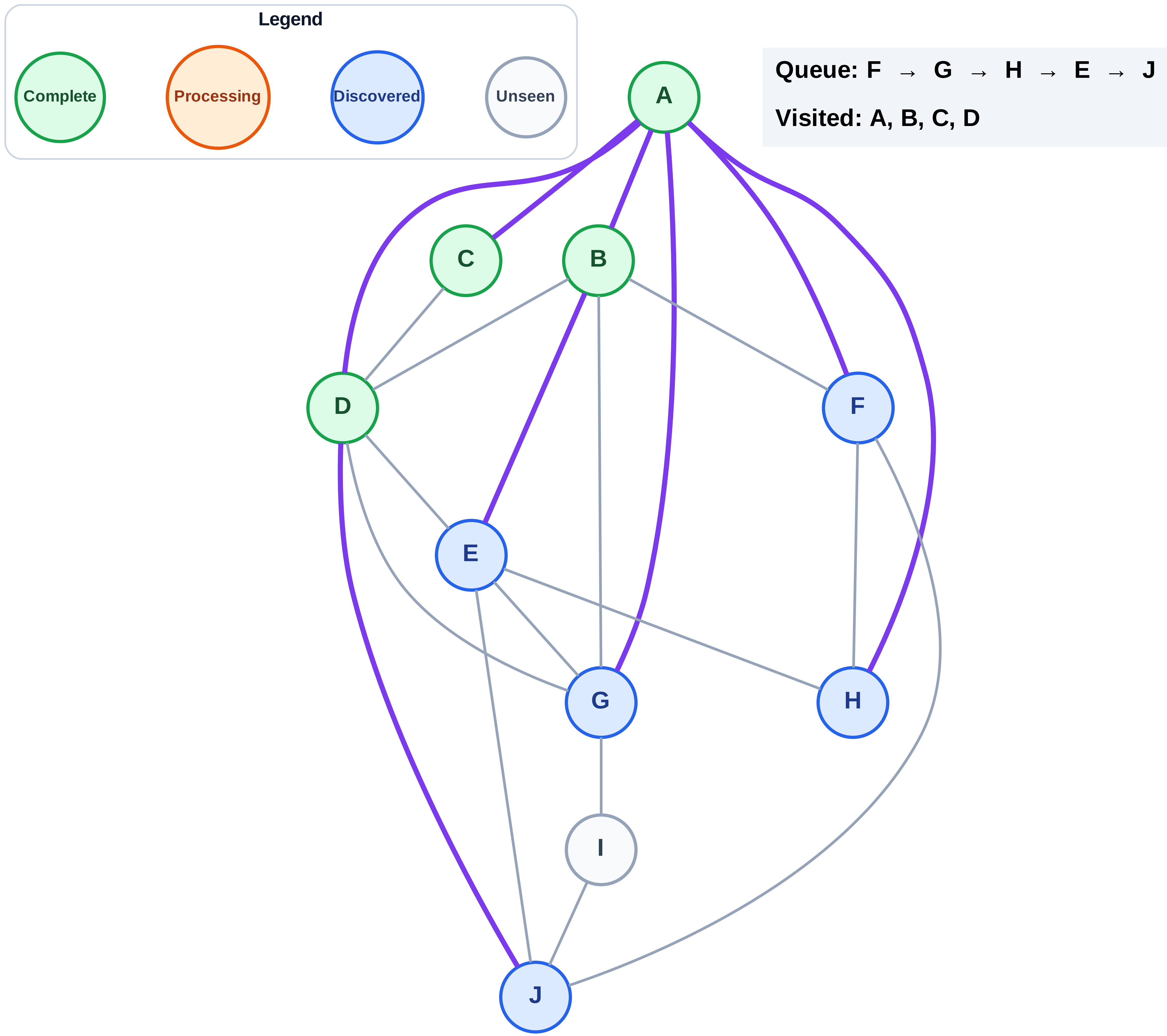
Step 7: No new neighbors from C




BFS Traversal
Step 8: Process node D
Legend
Complete
Processing
Discovered
Unseen
Queue: F → G → H → E
Visited: A, B, C
A
B
C
D
E
F
G
H
I
J

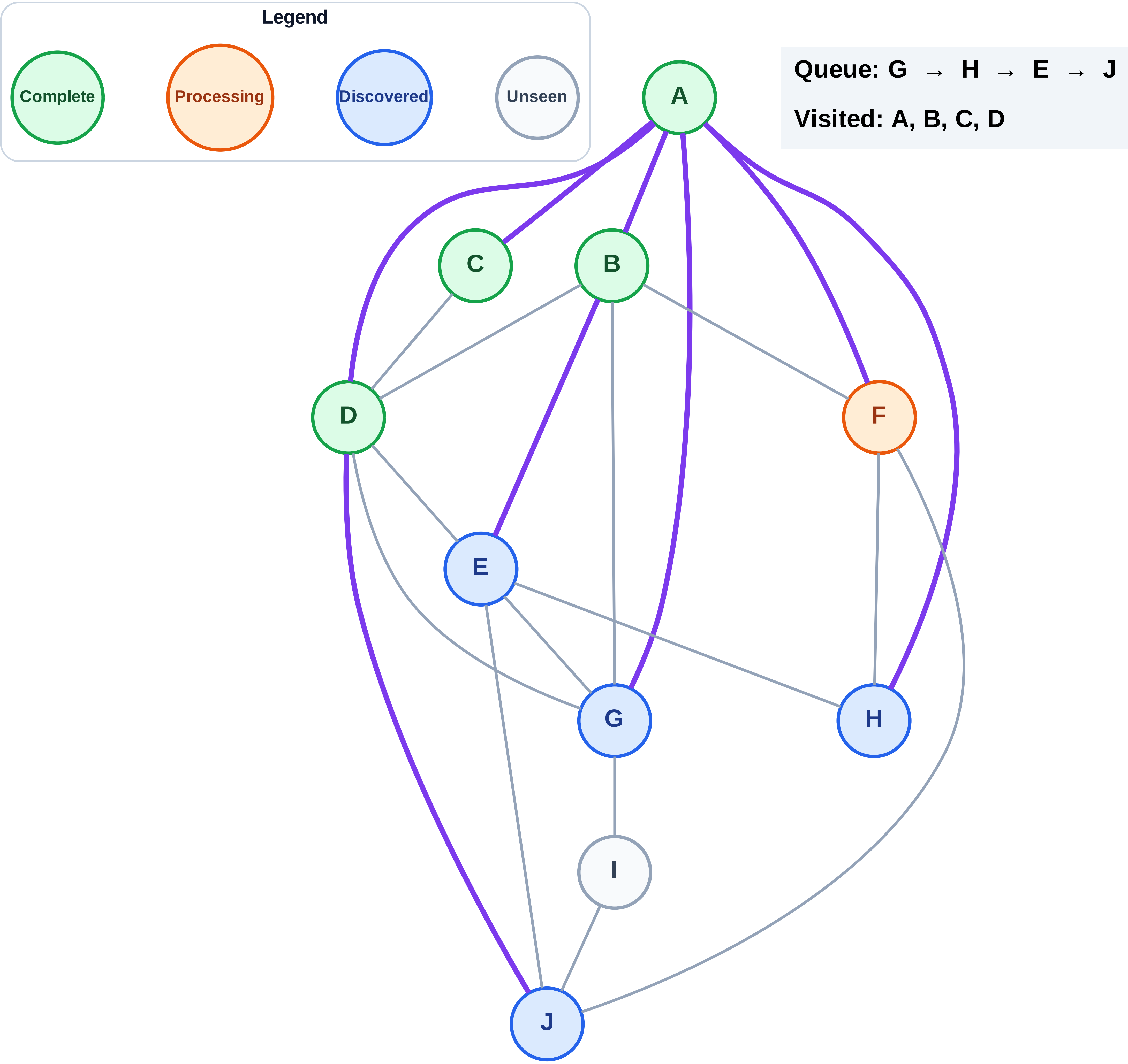
BFS Traversal

Step 9: Discover J from D



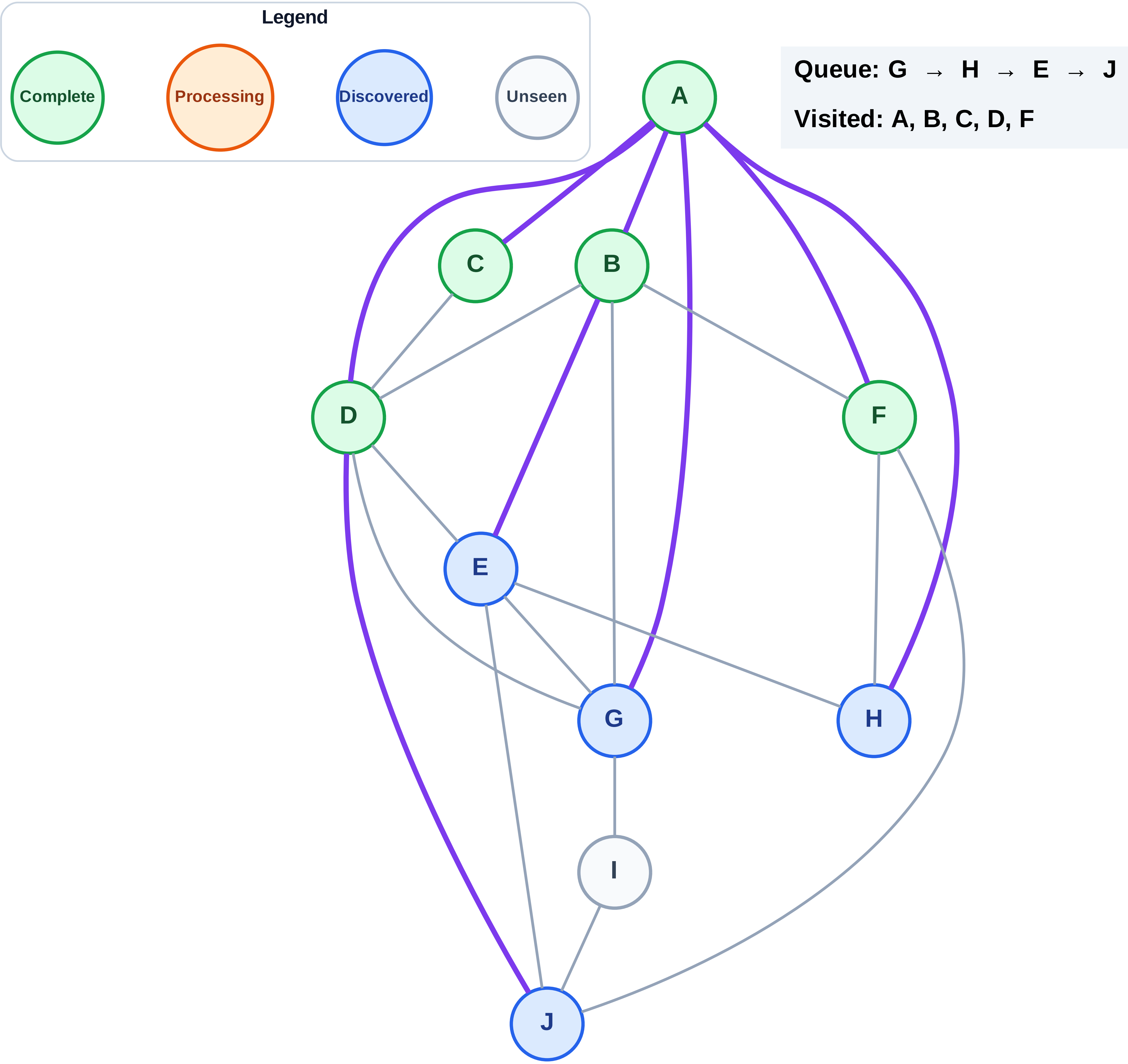
BFS Traversal

Step 10: Process node F



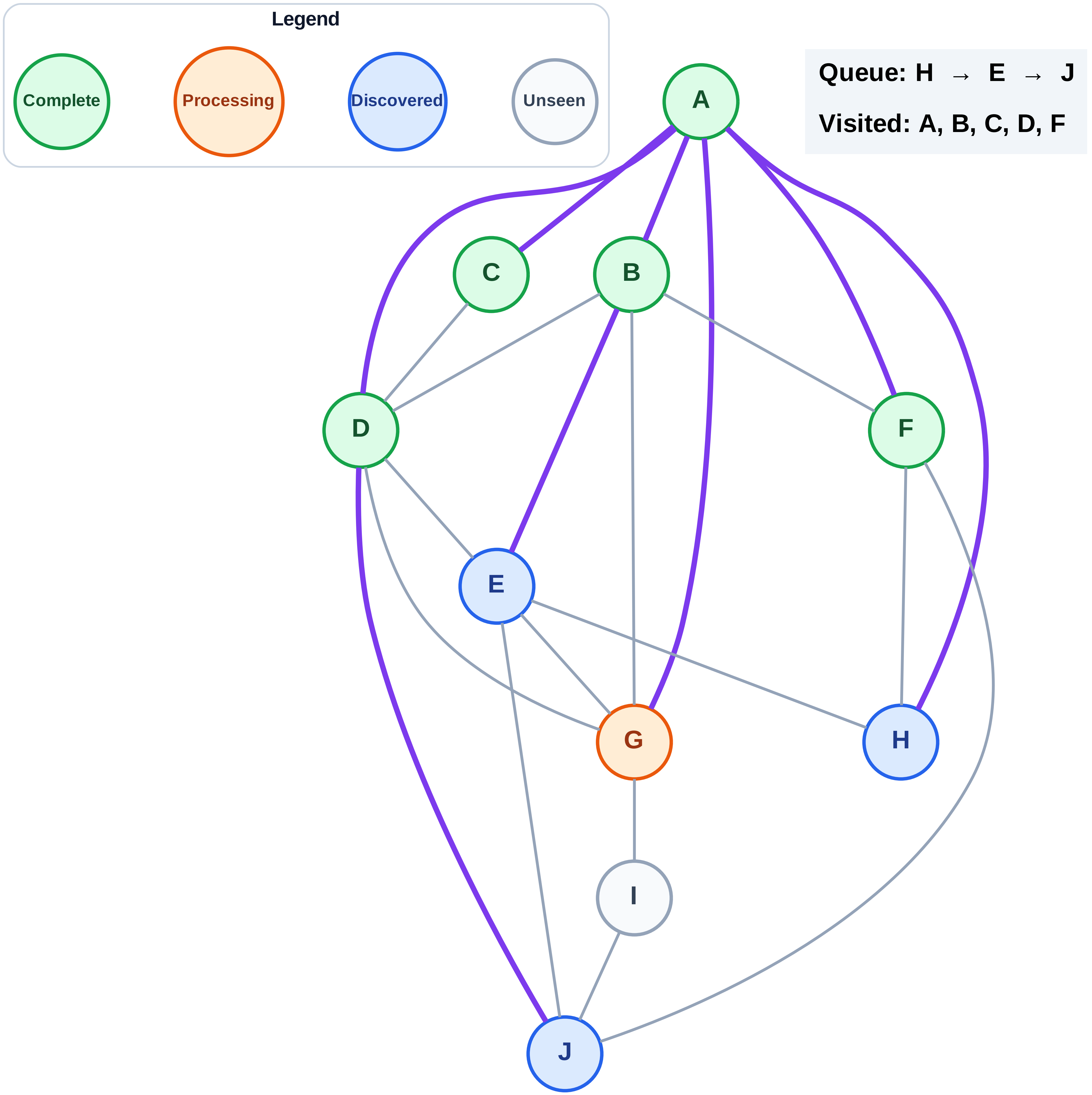
BFS Traversal

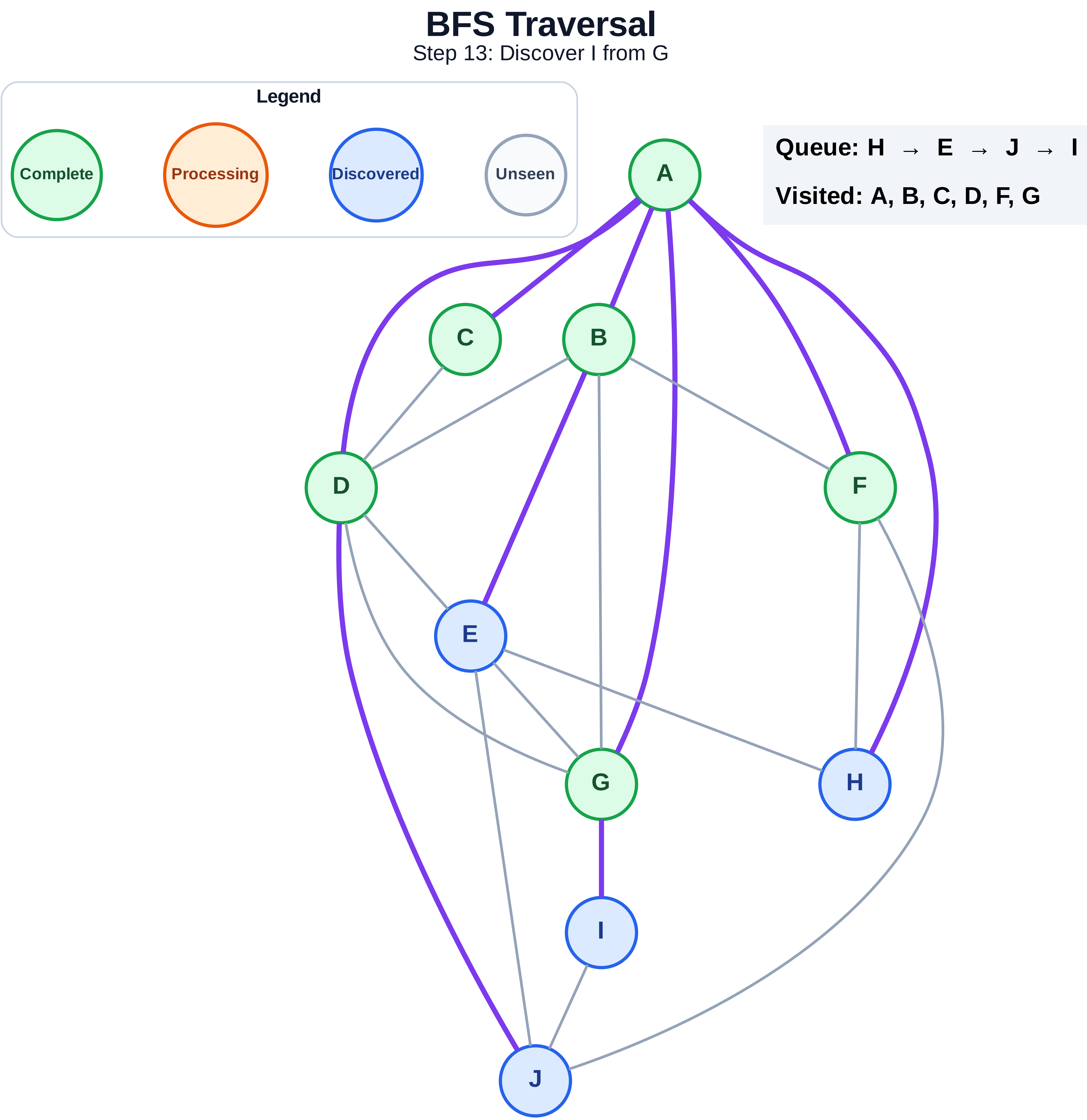
Step 11: No new neighbors from F



BFS Traversal

Step 12: Process node G

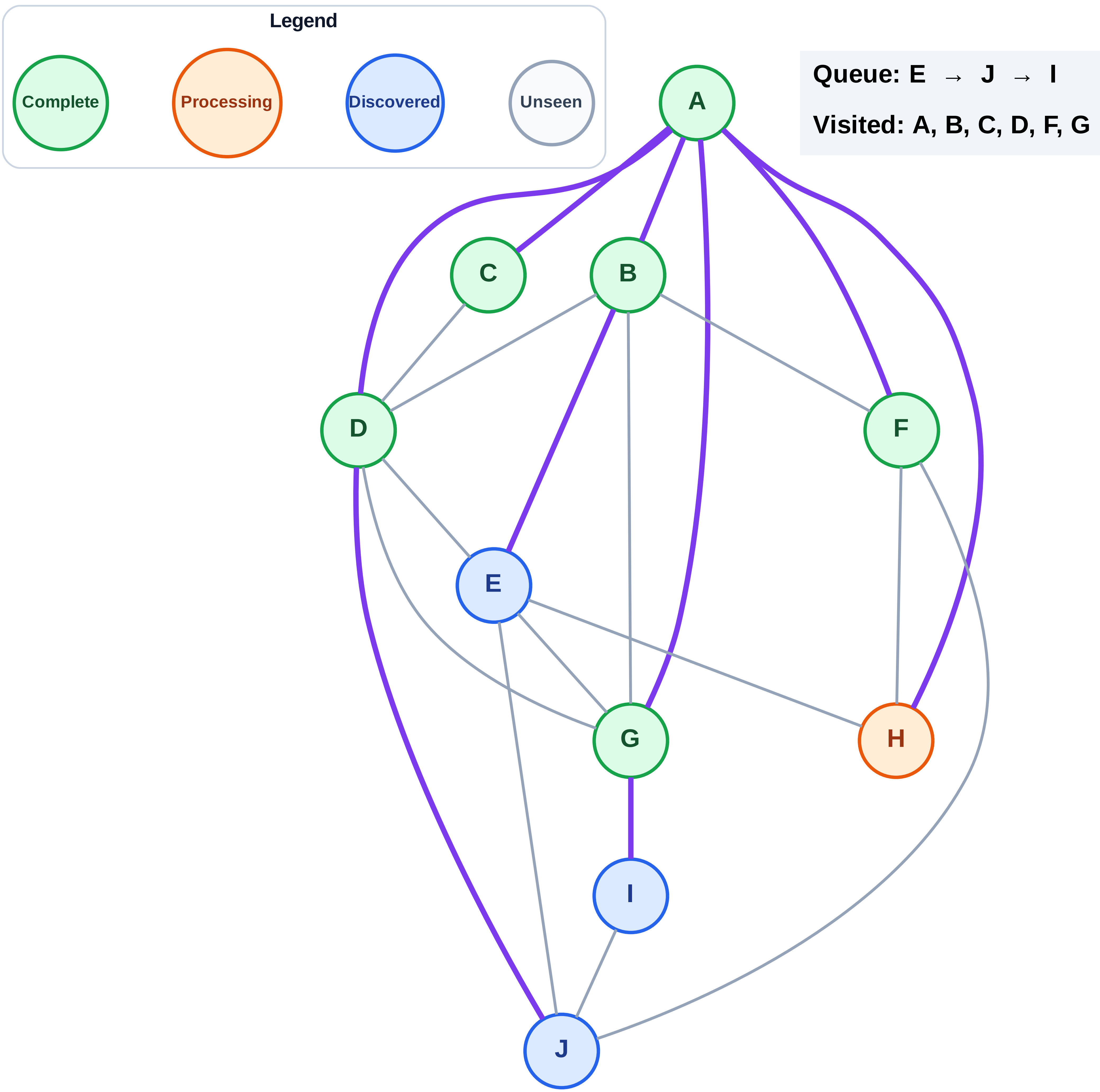




BFS Traversal
Step 13: Discover I from G
Legend
Complete
Processing
Discovered
Unseen
Queue: H → E → J → I
Visited: A, B, C, D, F, G
A
B
C
D
E
F
G
H
I
J

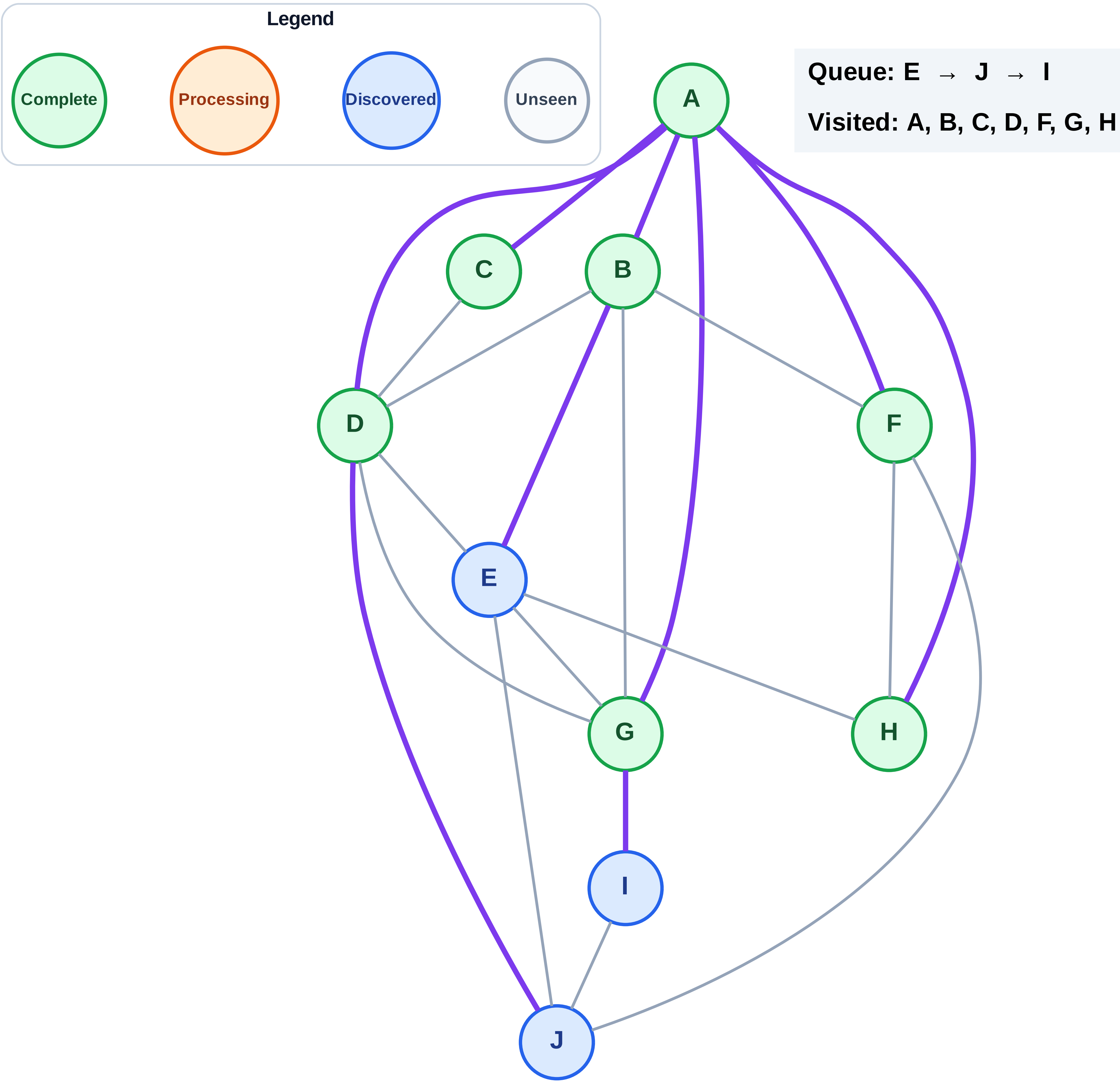
BFS Traversal

Step 14: Process node H



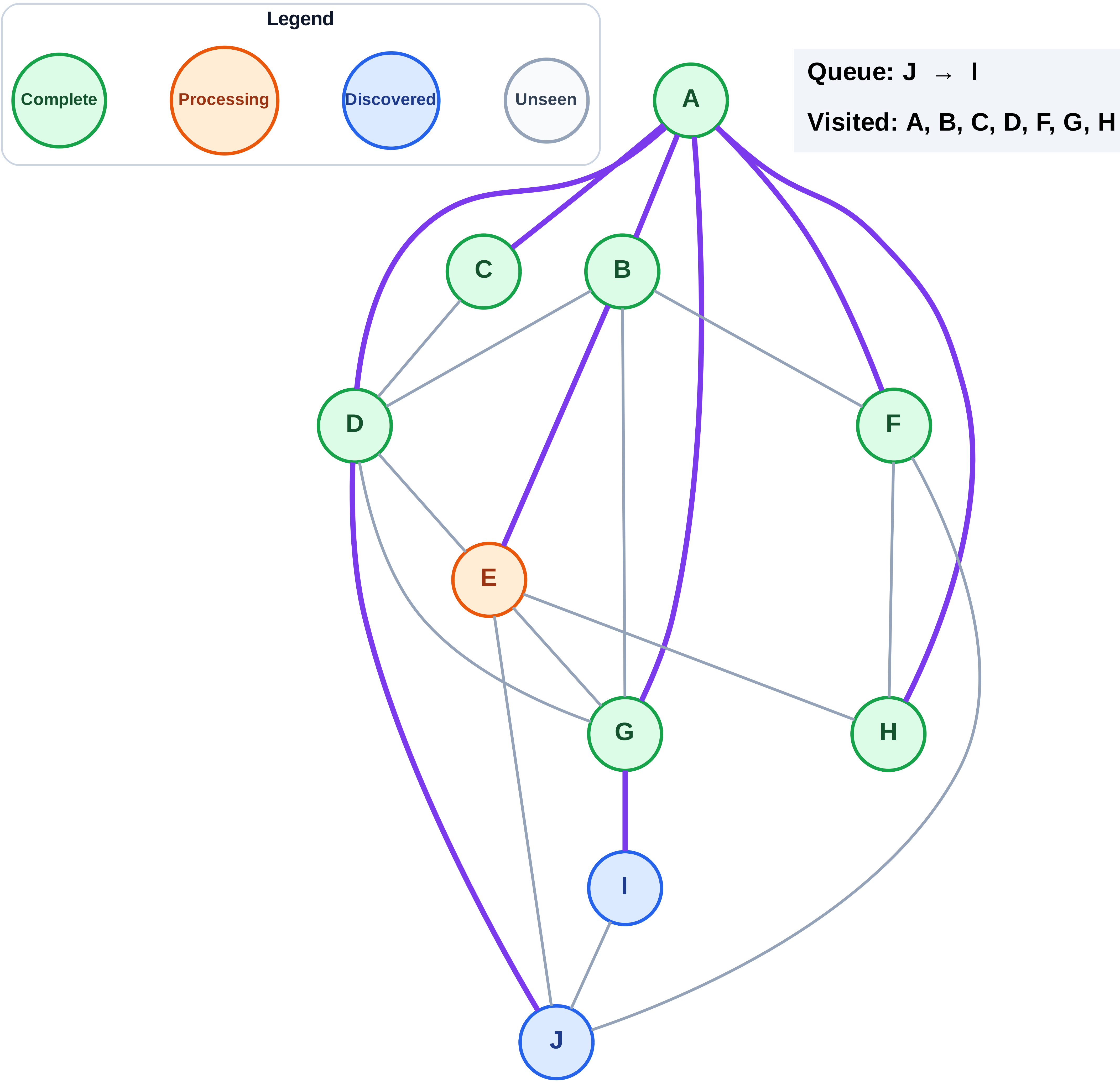
BFS Traversal

Step 15: No new neighbors from H



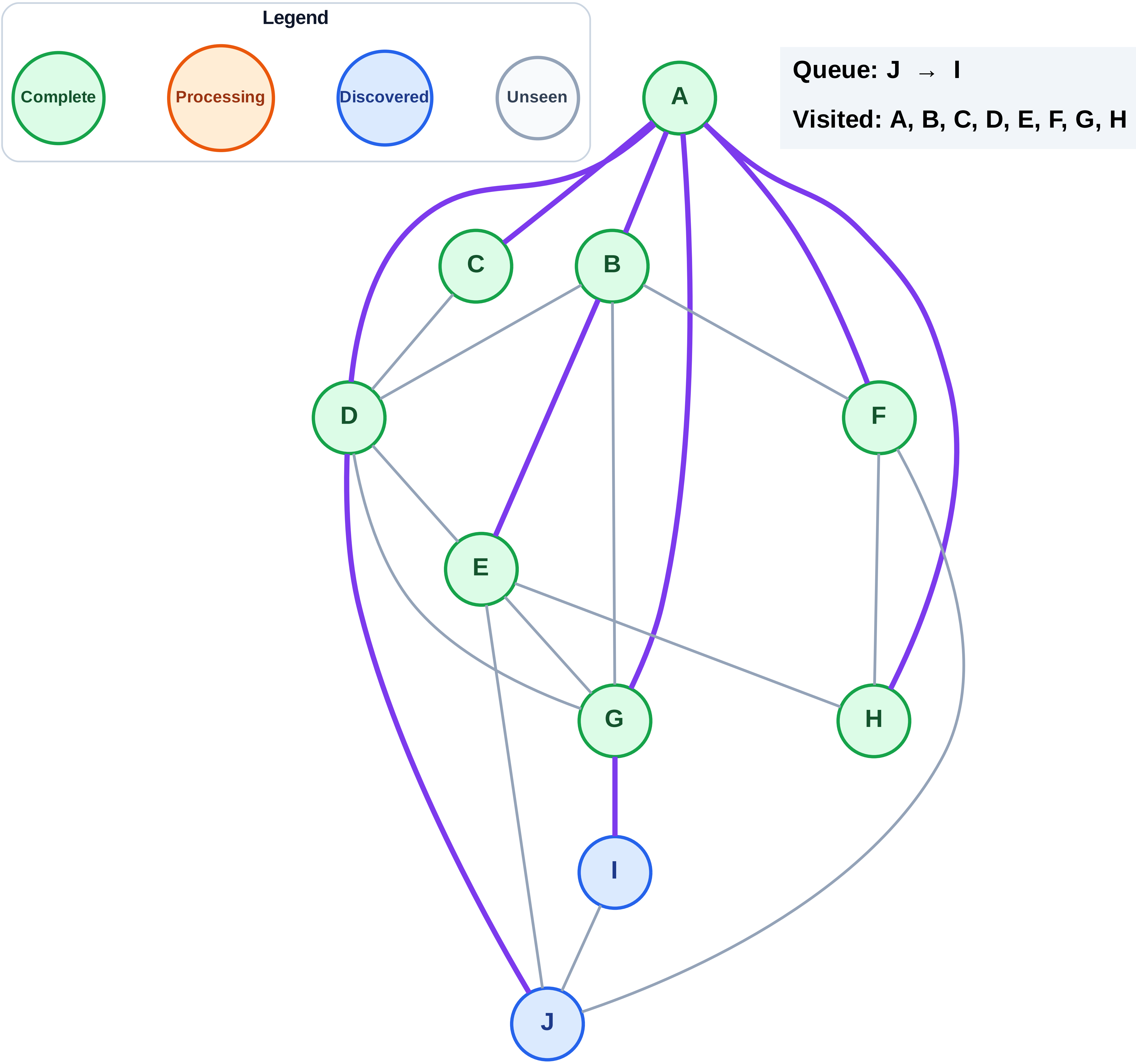
BFS Traversal

Step 16: Process node E



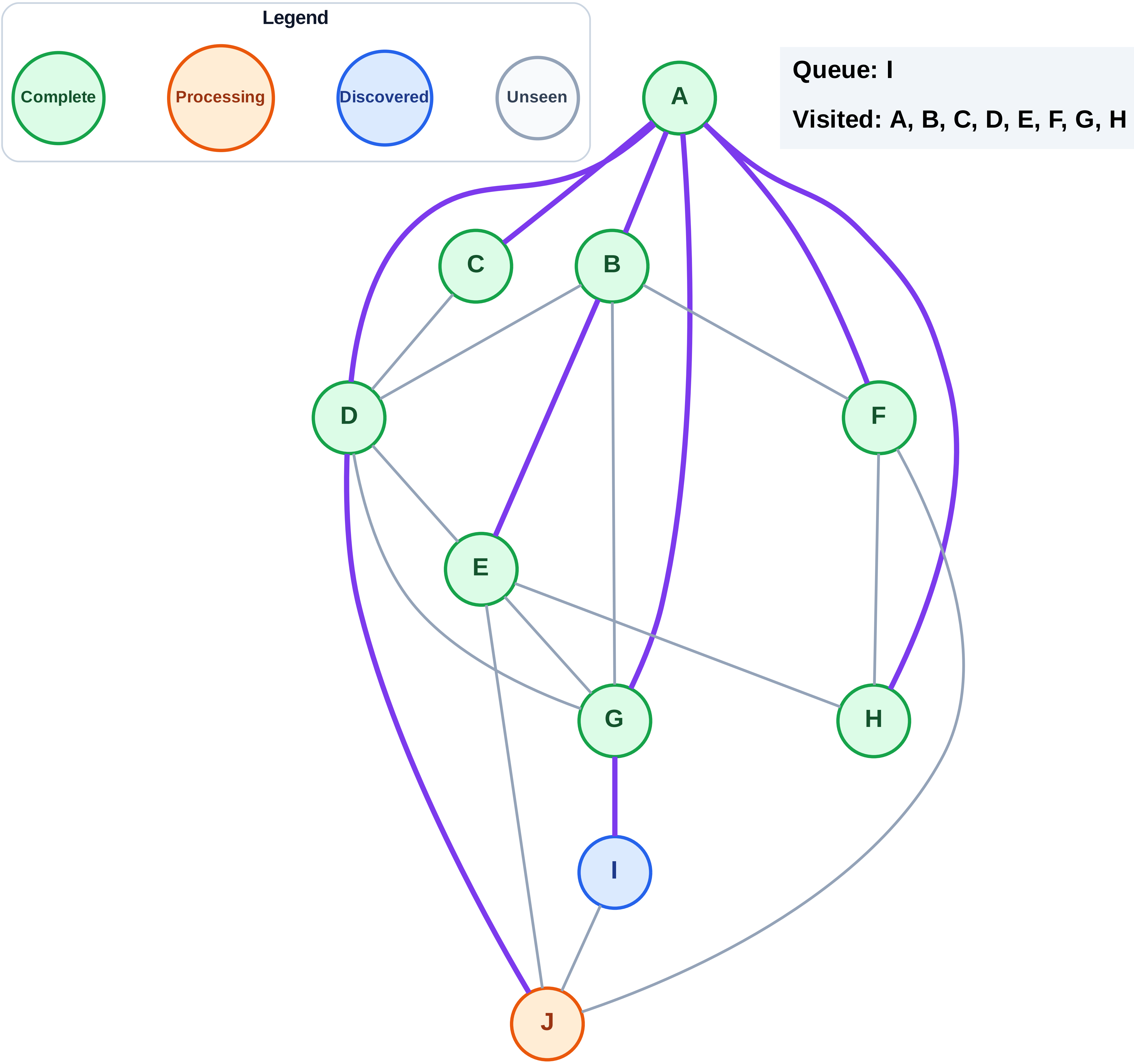
BFS Traversal

Step 17: No new neighbors from E



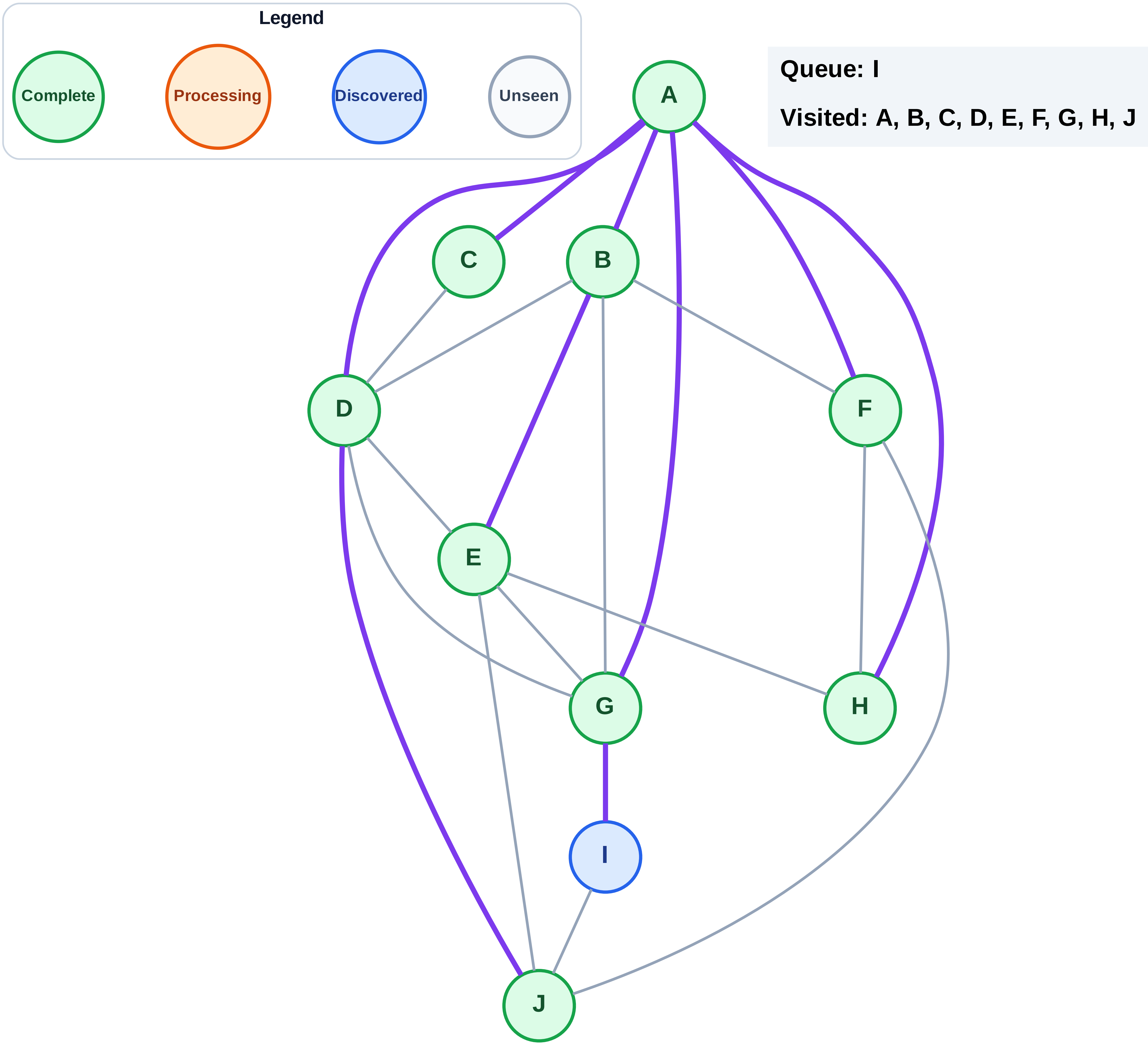
BFS Traversal

Step 18: Process node J



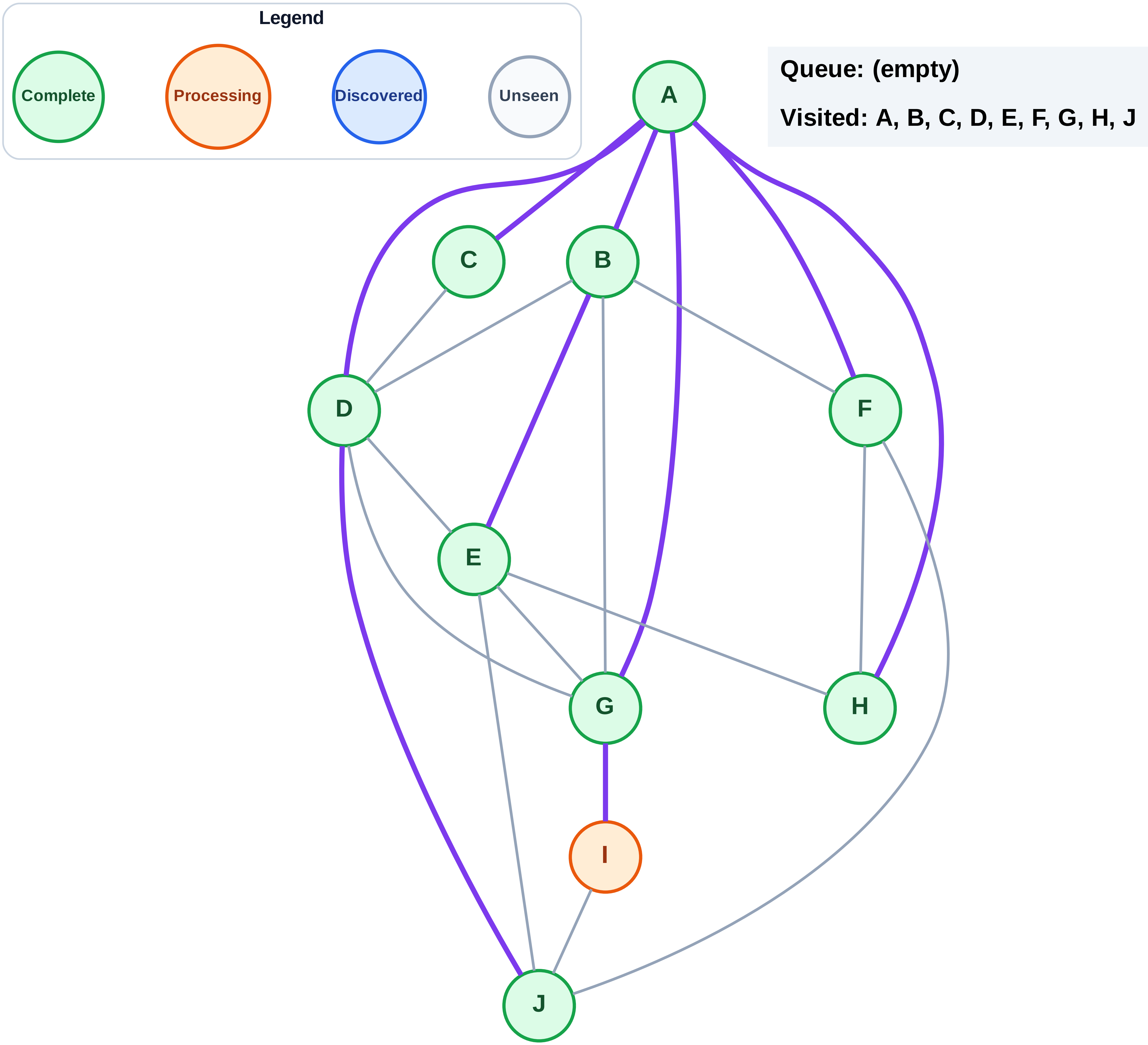
BFS Traversal

Step 19: No new neighbors from J



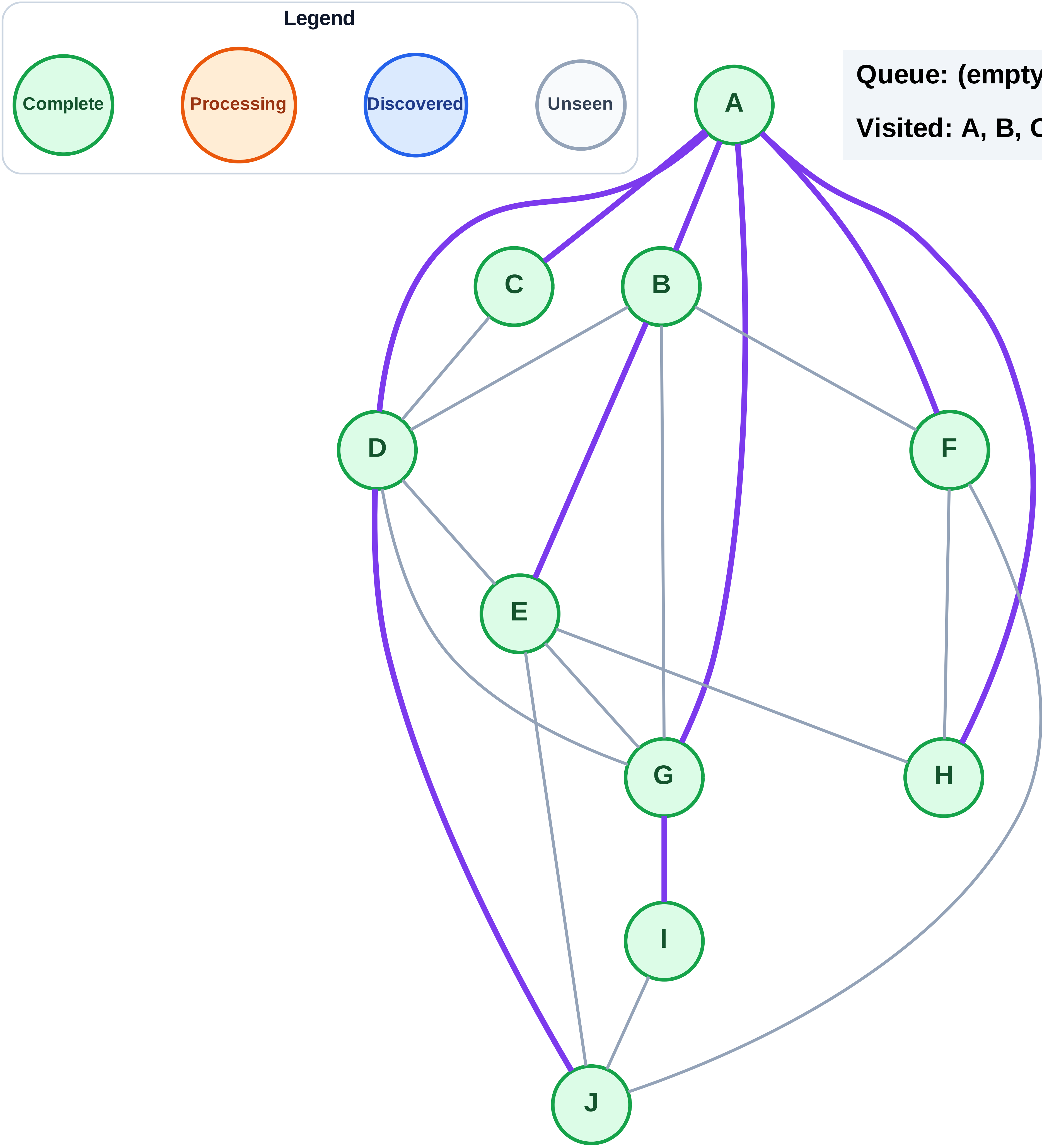
BFS Traversal

Step 20: Process node I



BFS Traversal

Step 21: No new neighbors from I

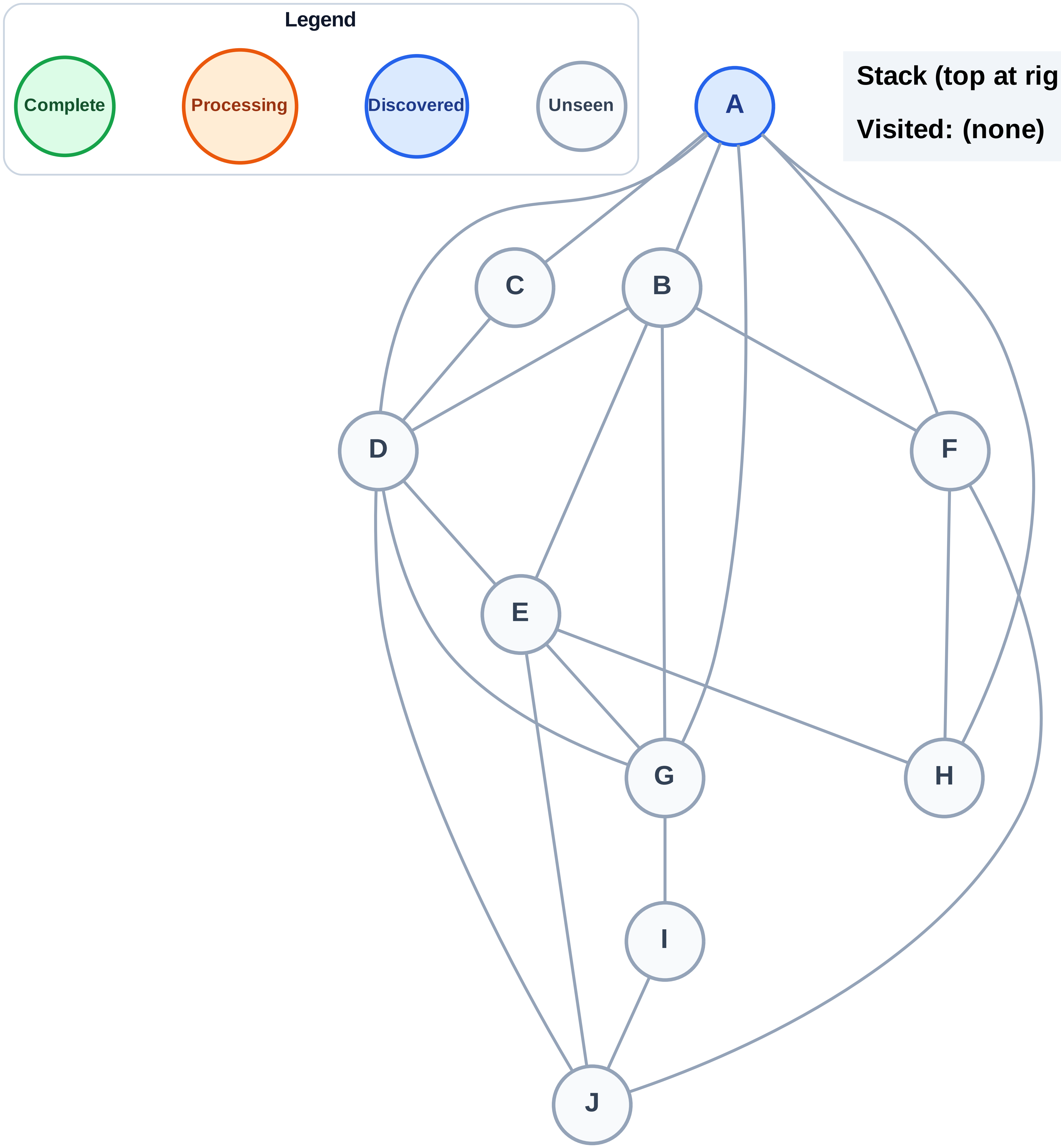


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at A

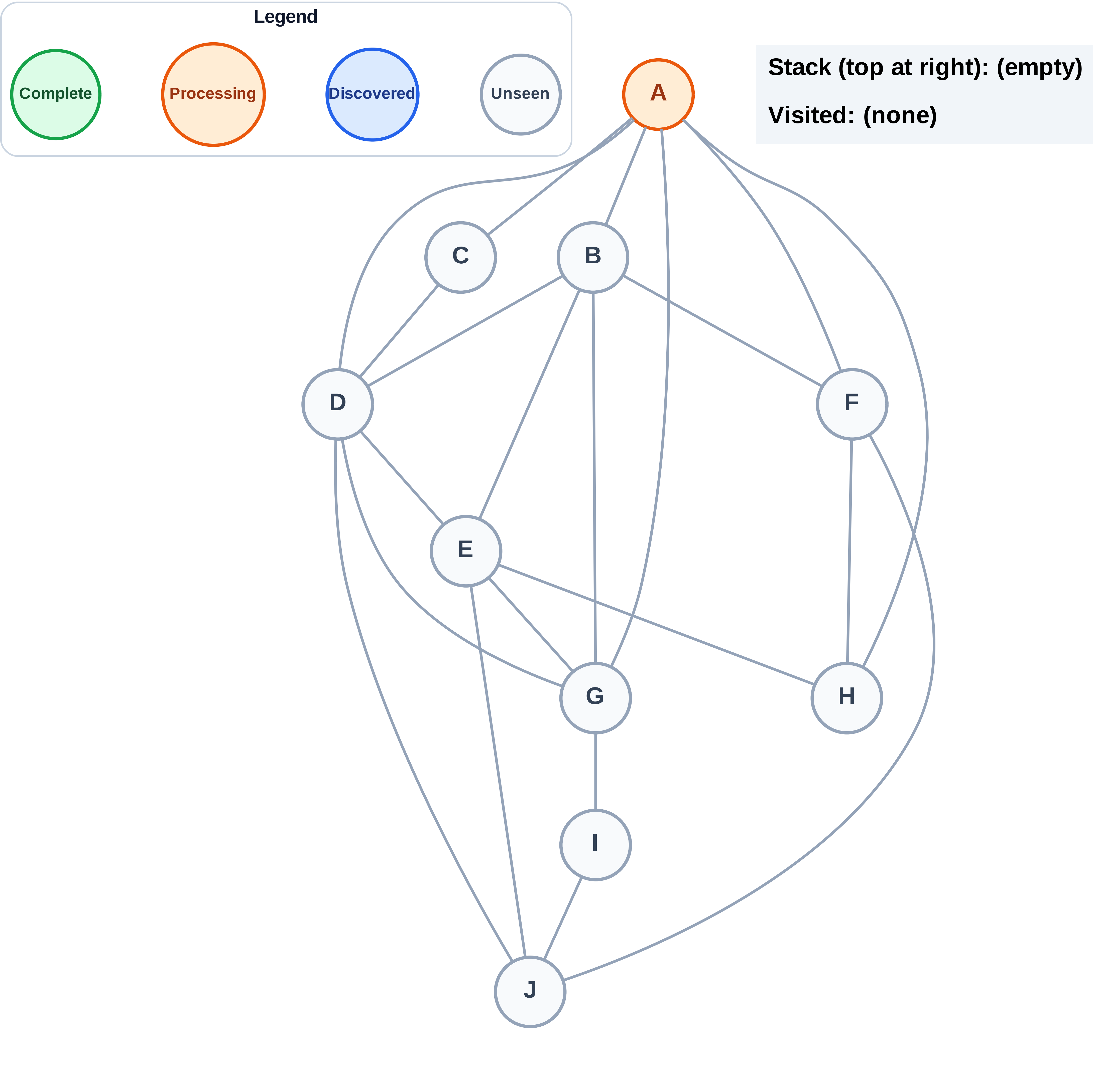


Stack (top at right): A

Visited: (none)

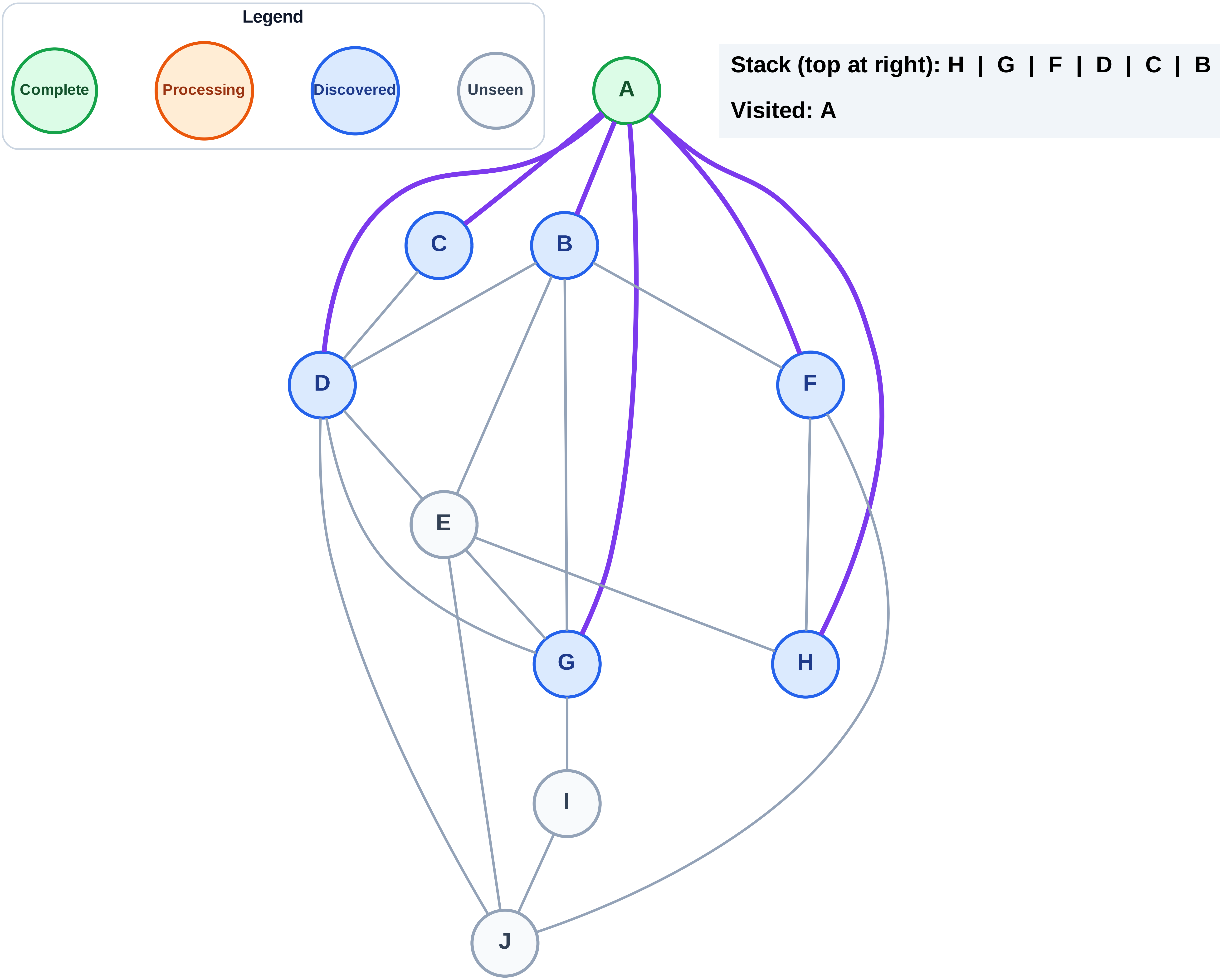
DFS Traversal

Step 2: Process node A



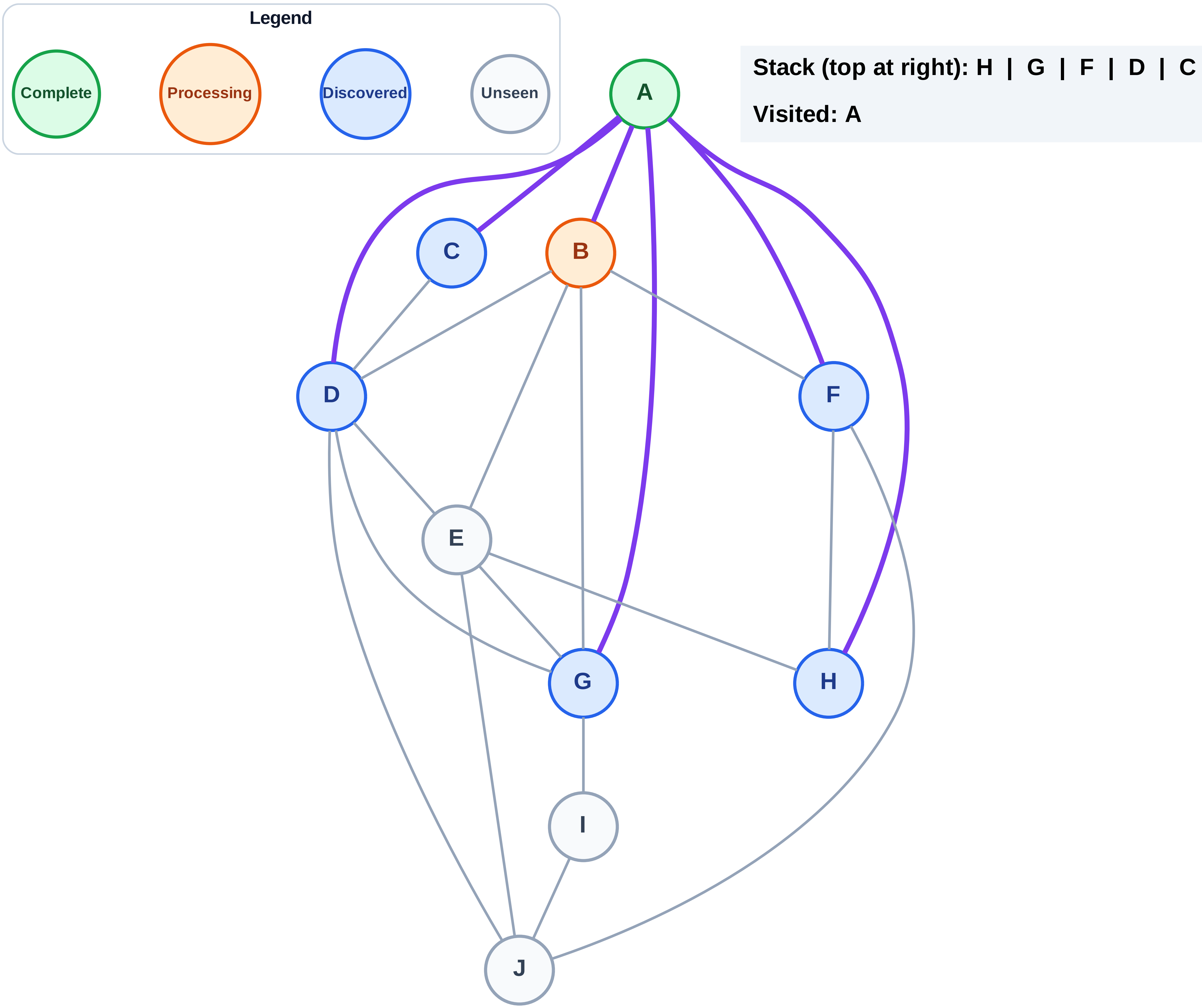
DFS Traversal

Step 3: Discover H, G, F, D, C, B from A



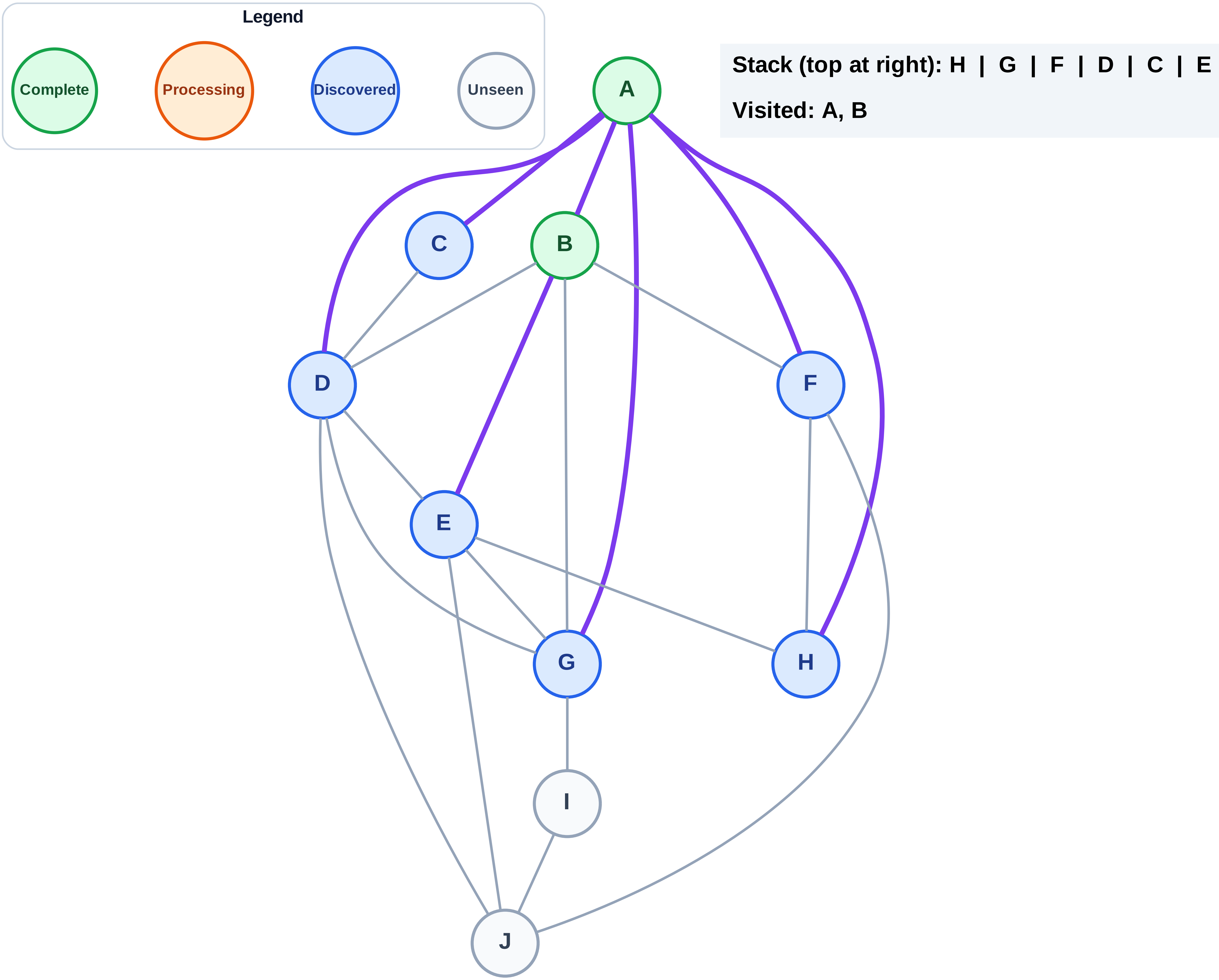
DFS Traversal

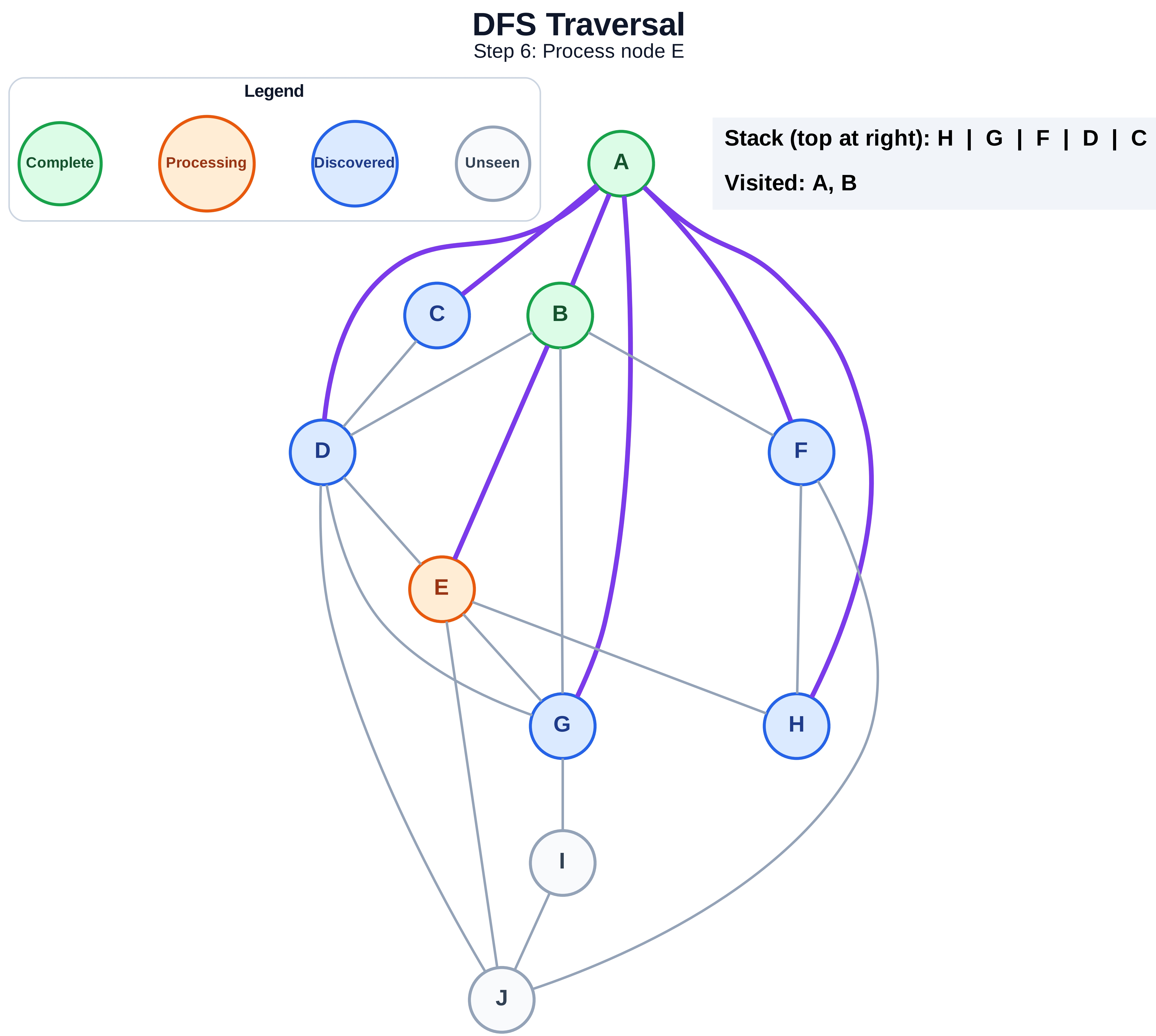
Step 4: Process node B



DFS Traversal

Step 5: Discover E from B

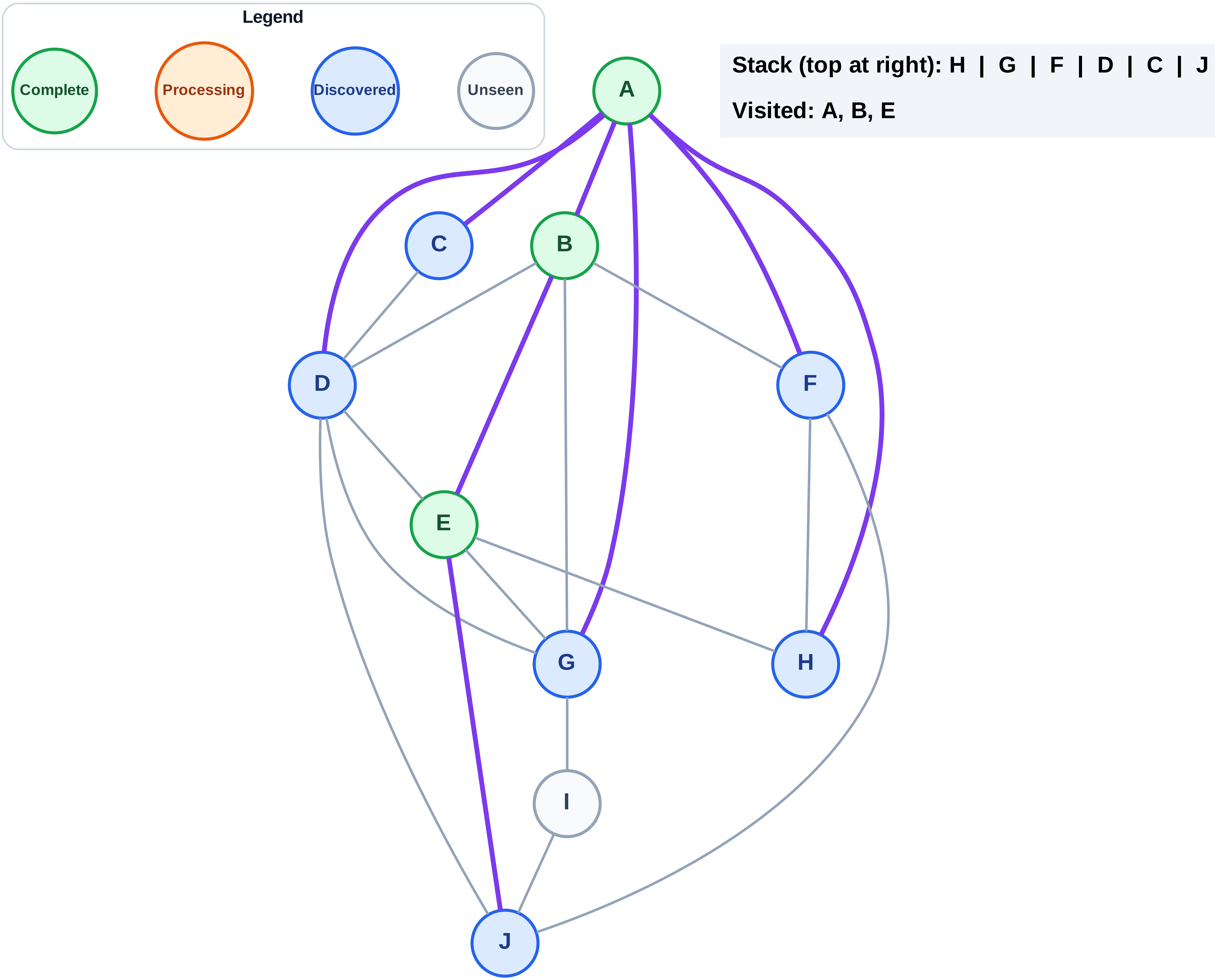




DFS Traversal
Step 6: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | G | F | D | C
Visited: A, B
A
B
C
D
E
F
G
H
I
J

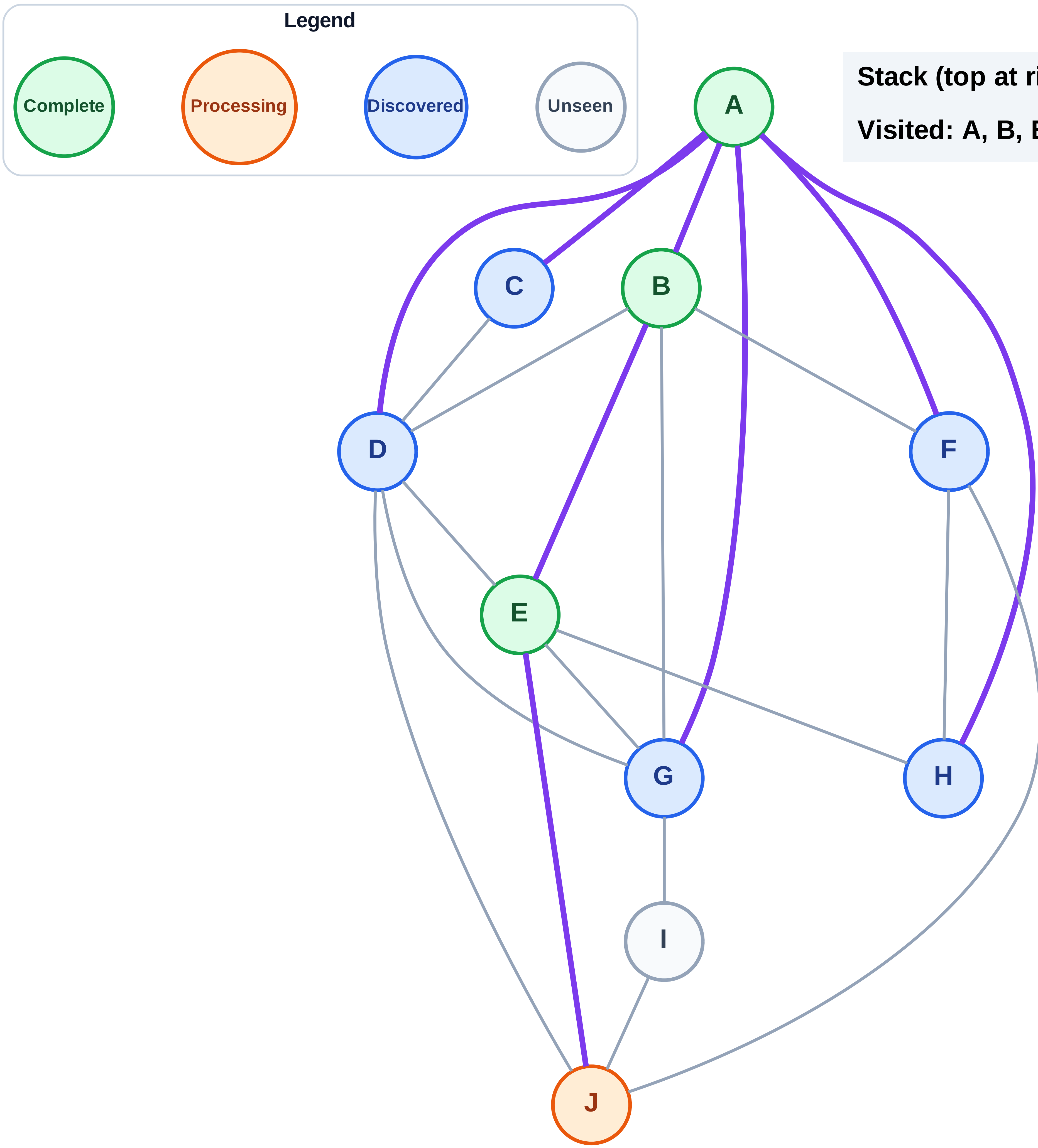
DFS Traversal

Step 7: Discover J from E



DFS Traversal

Step 8: Process node J

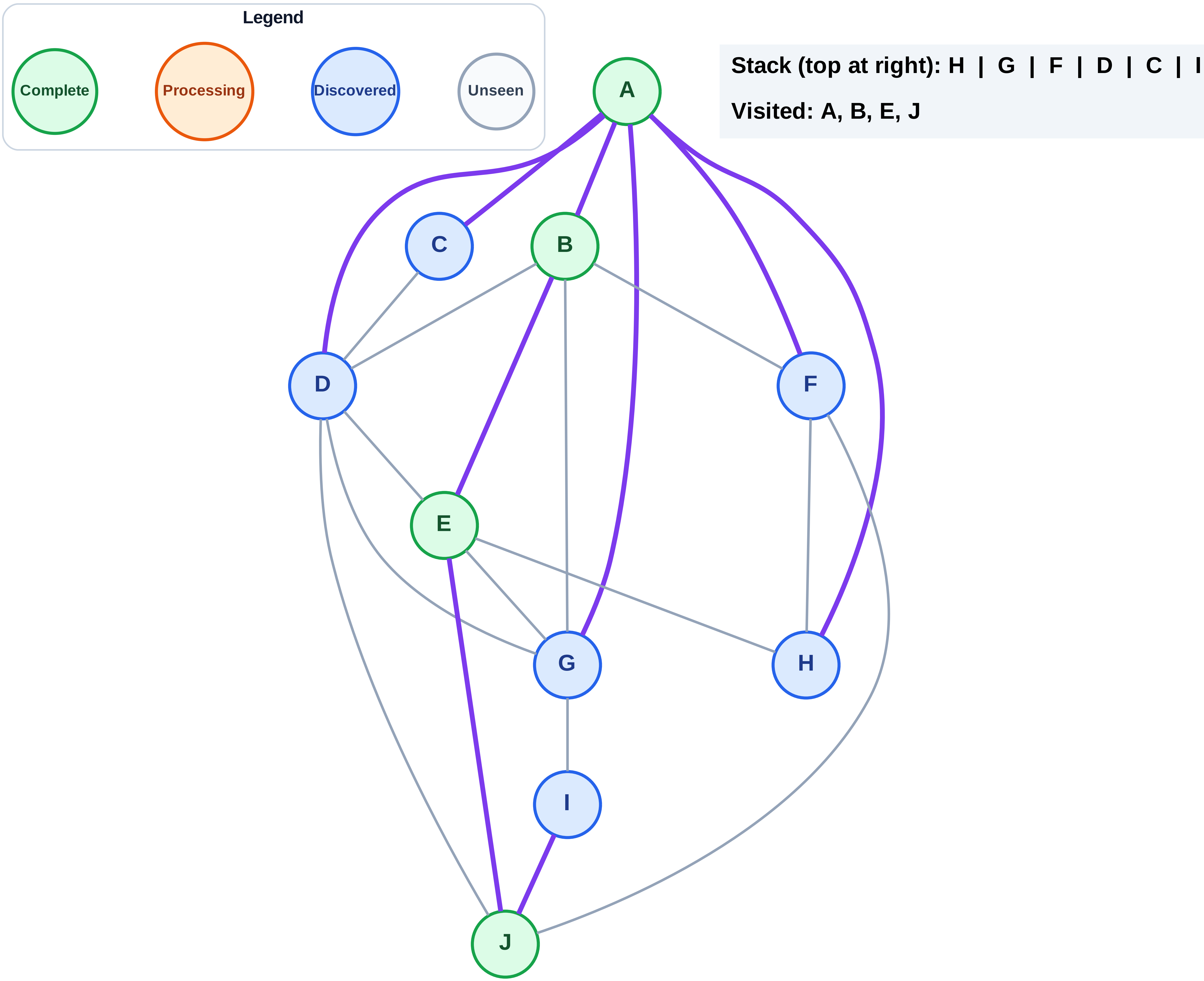


Stack (top at right): H | G | F | D | C

Visited: A, B, E

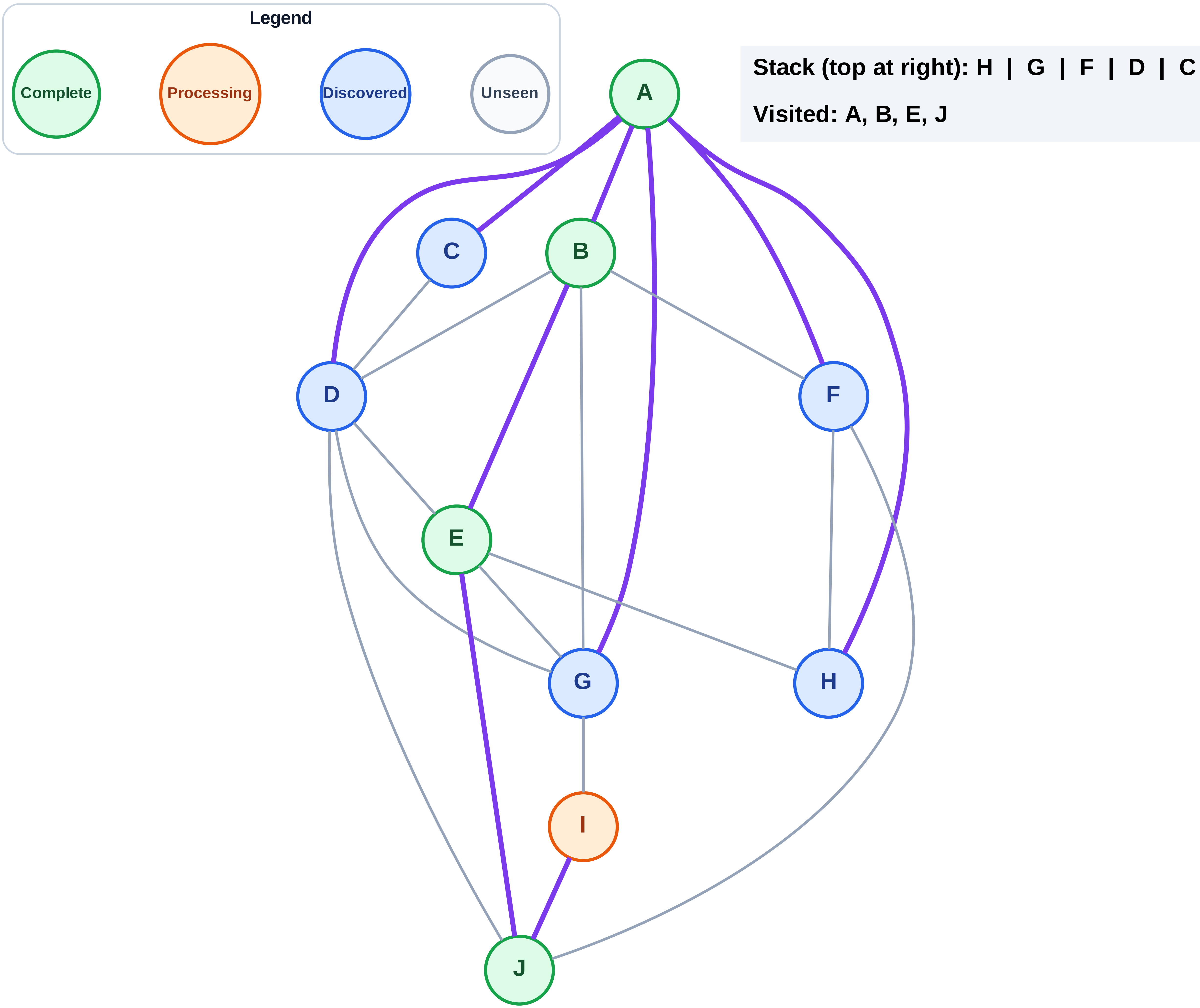
DFS Traversal

Step 9: Discover I from J



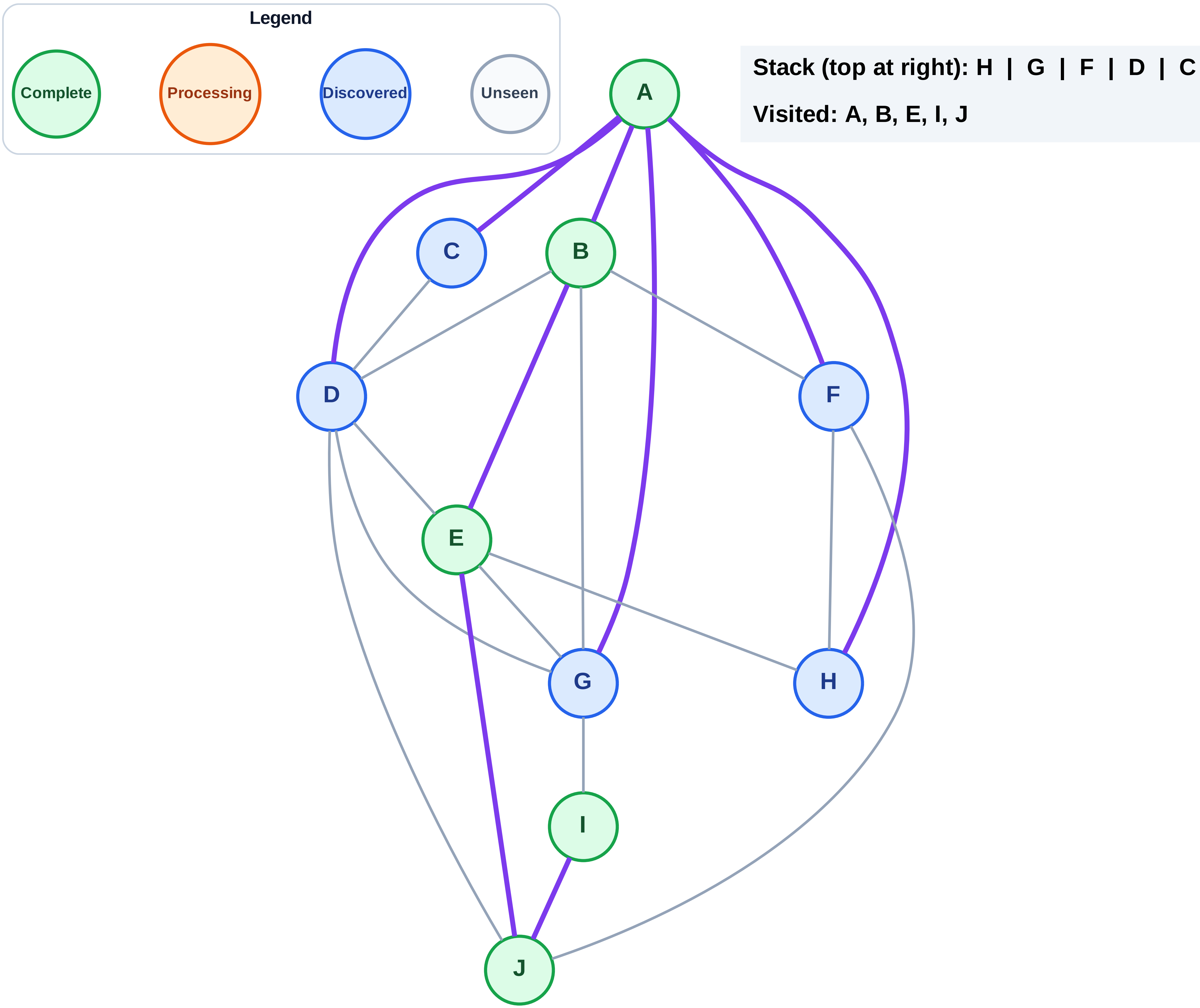
DFS Traversal

Step 10: Process node I



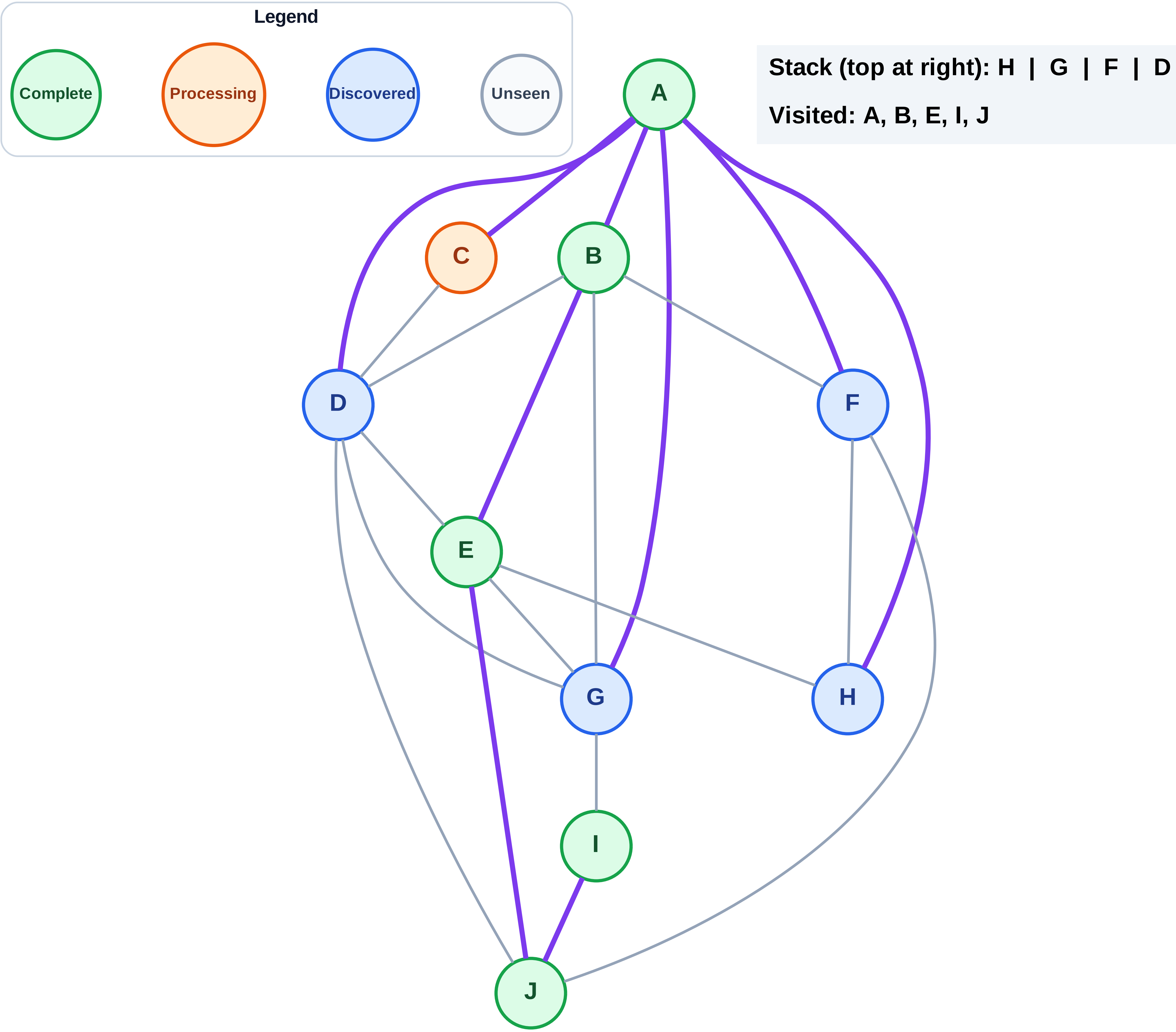
DFS Traversal

Step 11: No new neighbors from I



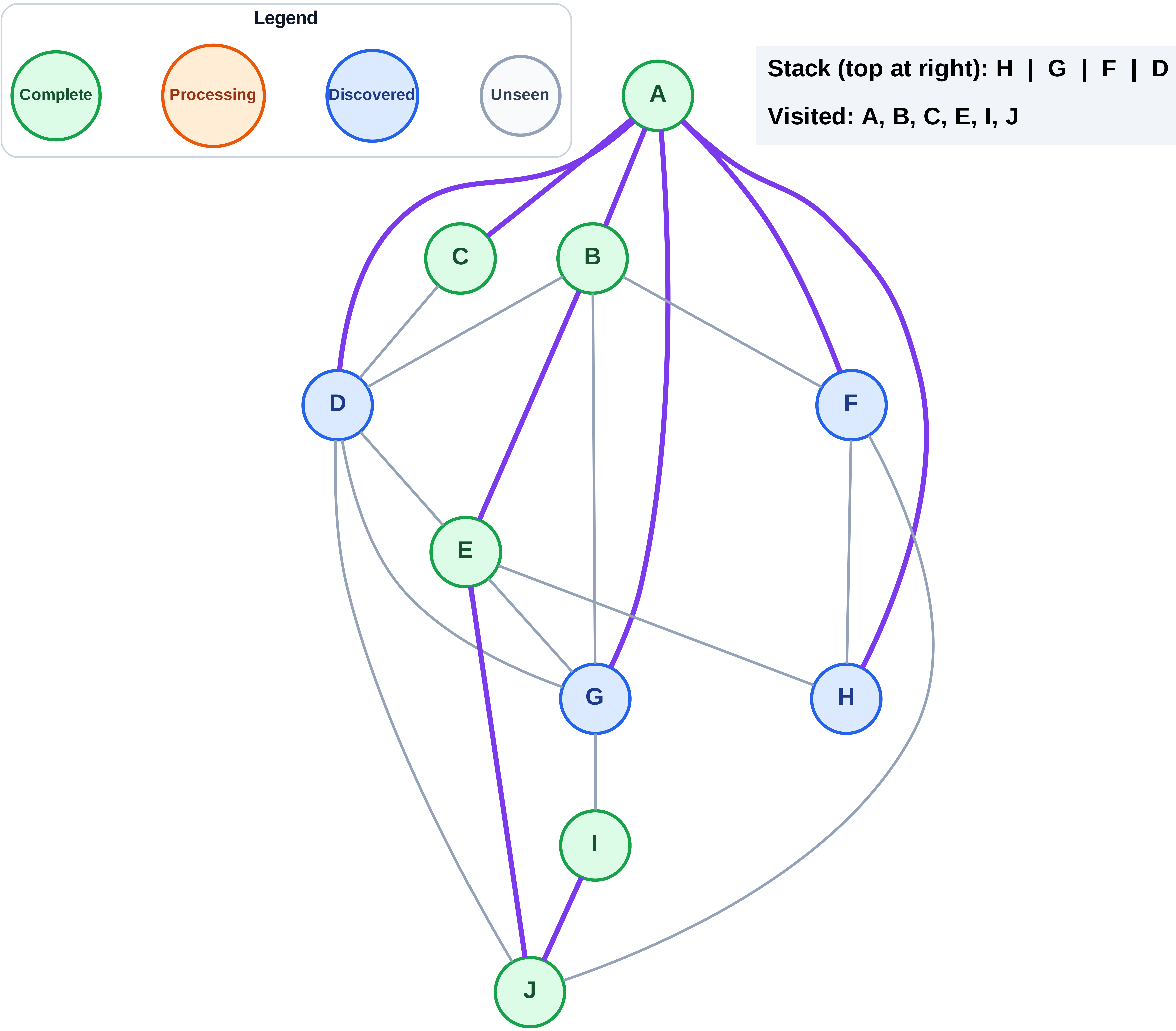
DFS Traversal

Step 12: Process node C



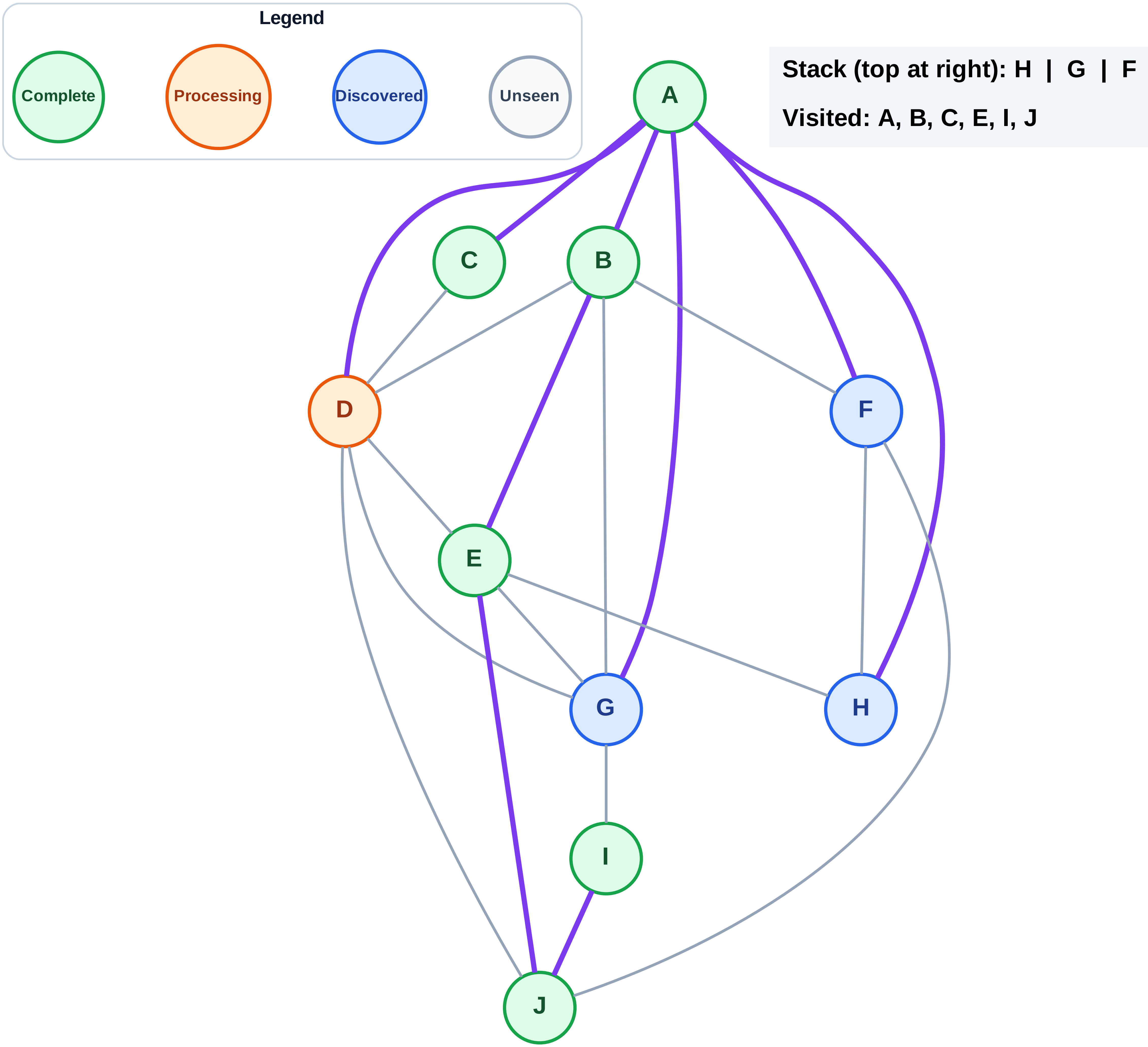
DFS Traversal

Step 13: No new neighbors from C



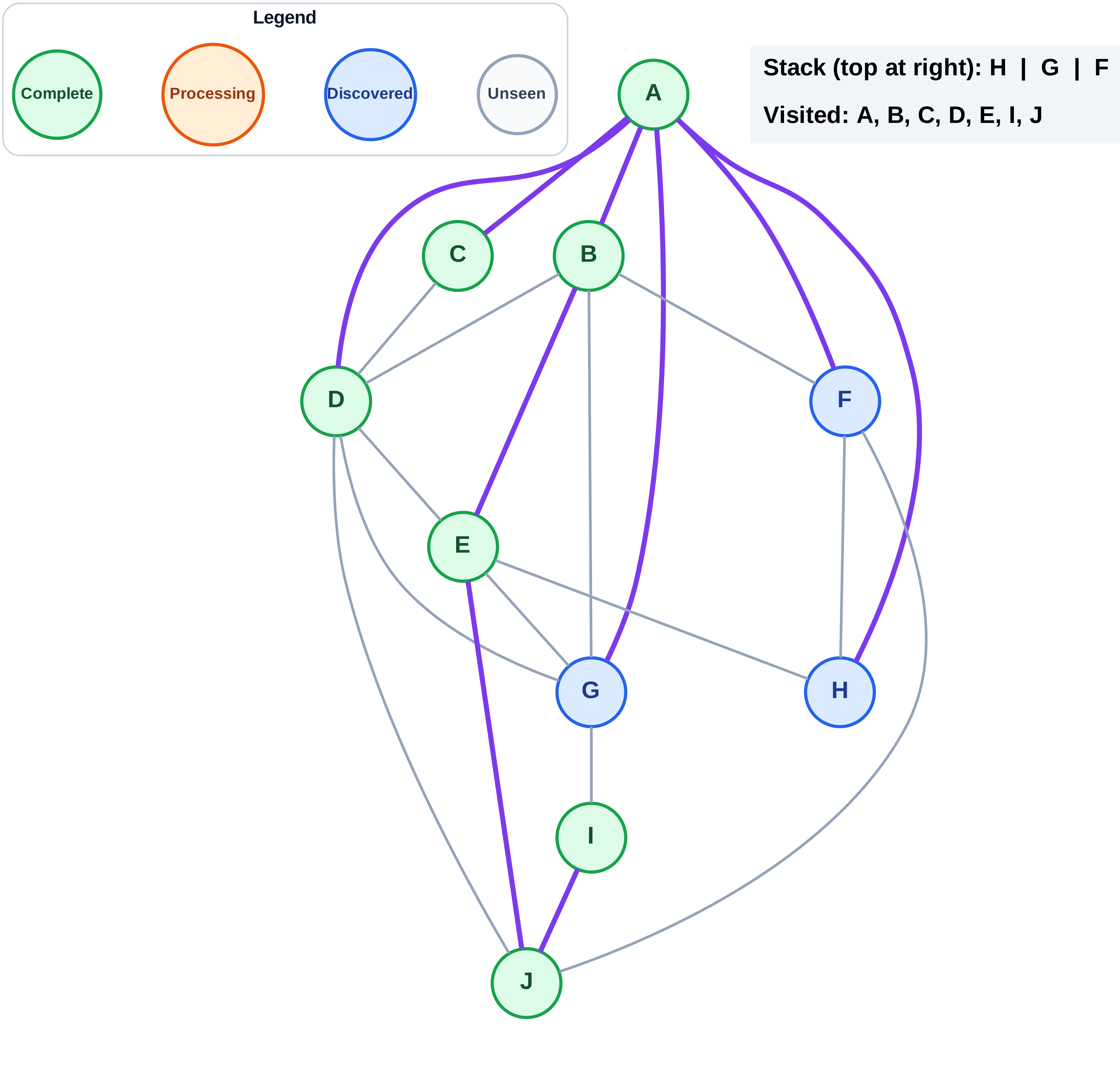
DFS Traversal

Step 14: Process node D



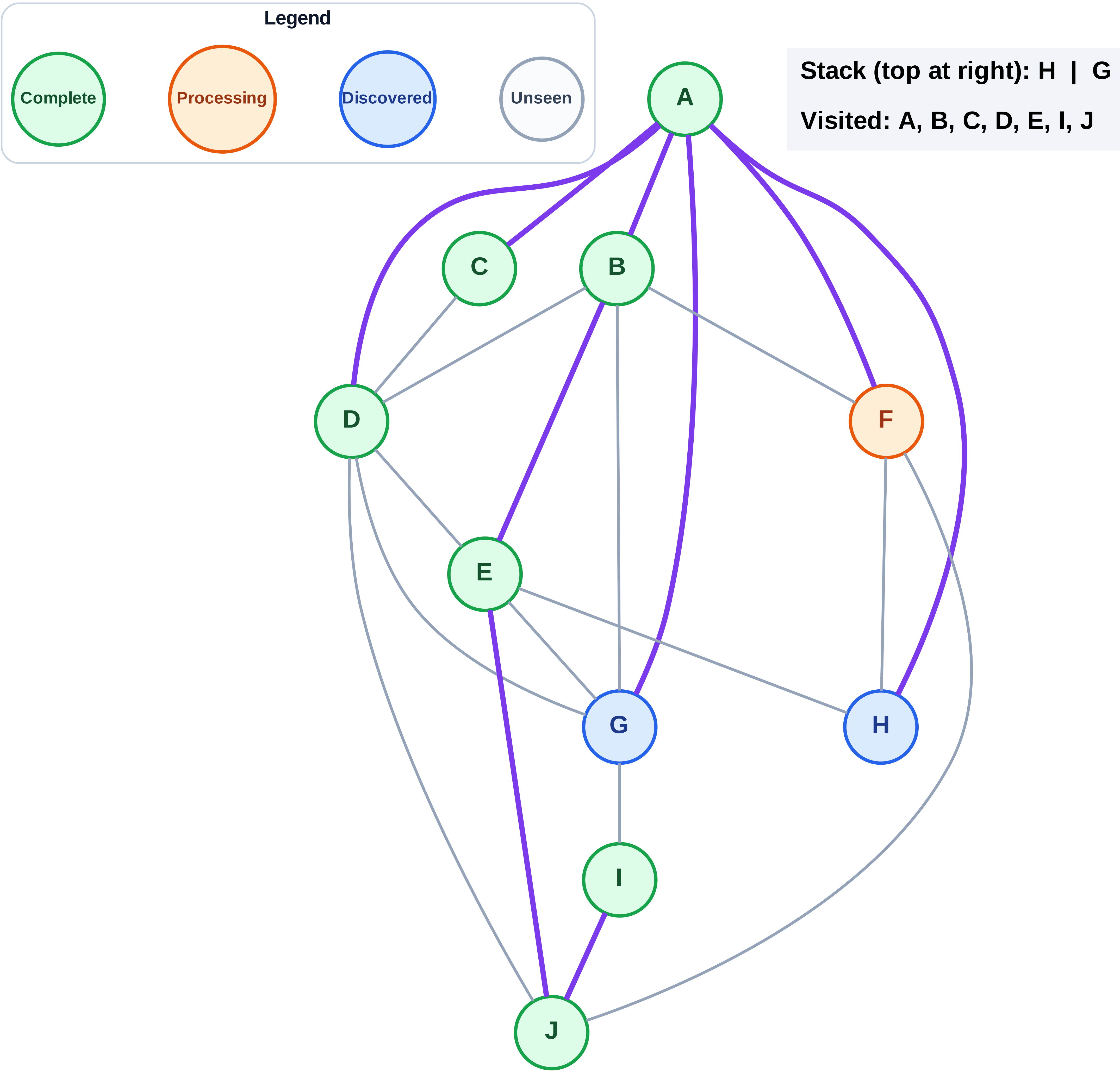
DFS Traversal

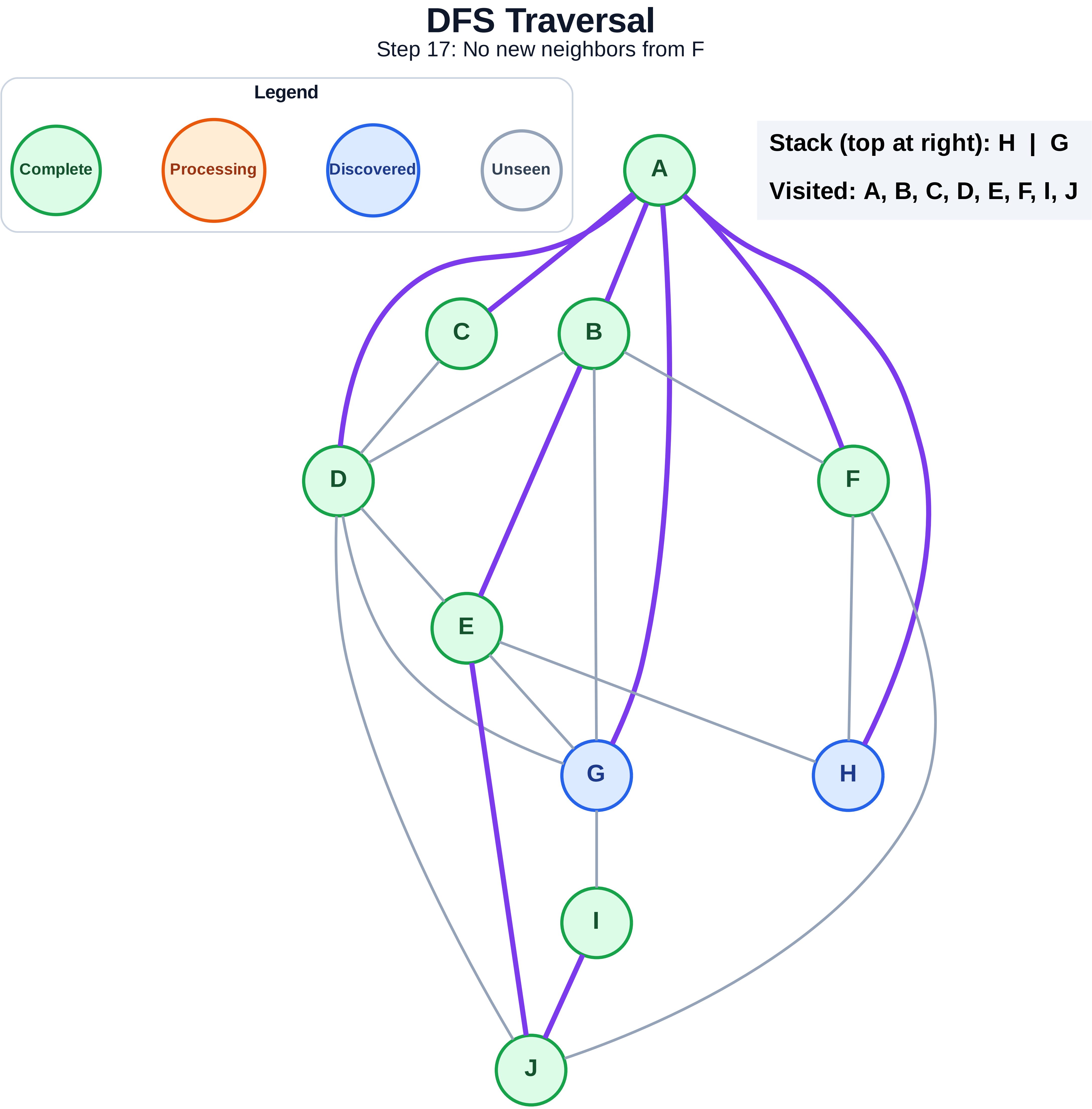
Step 15: No new neighbors from D

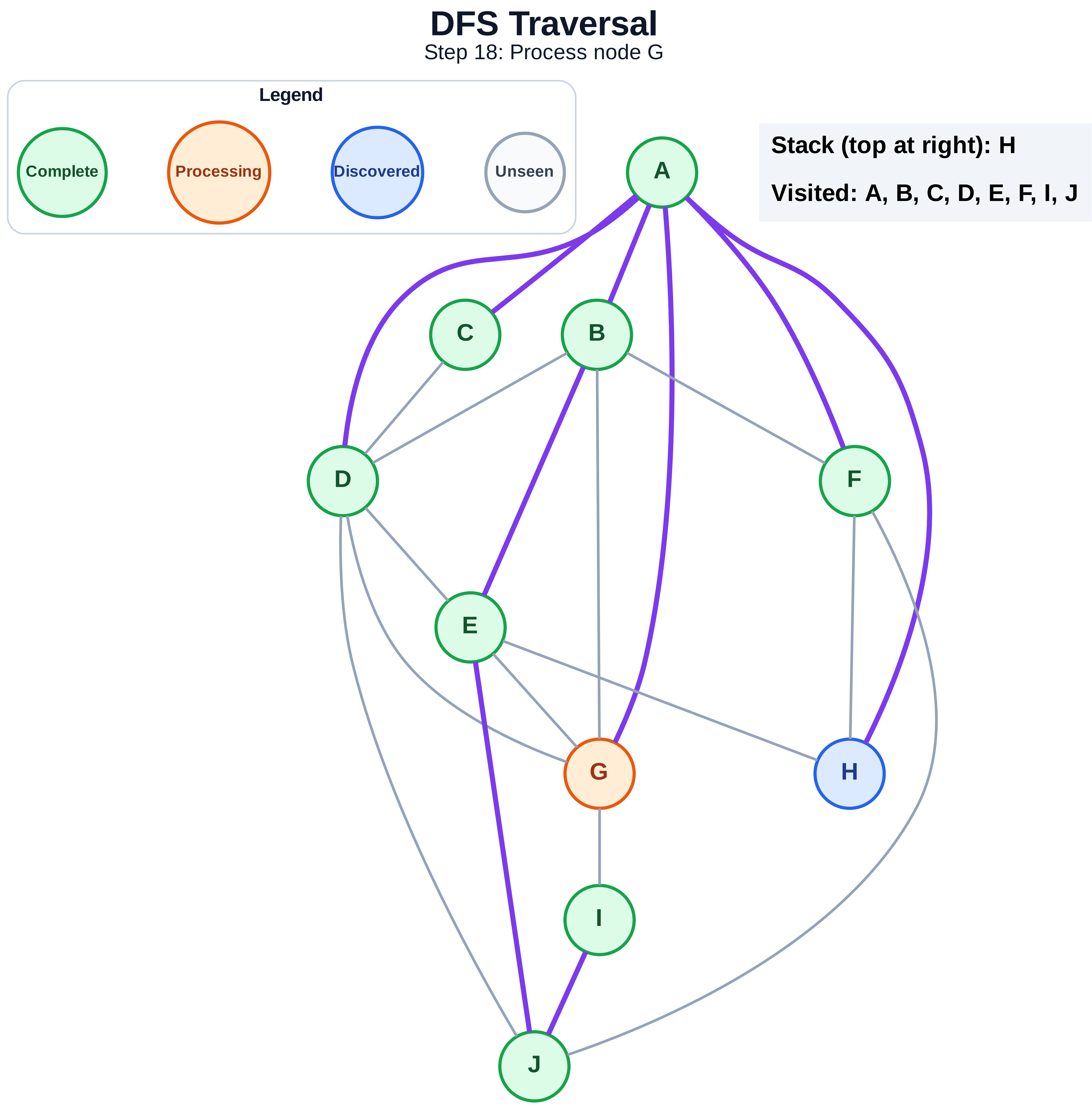


DFS Traversal

Step 16: Process node F

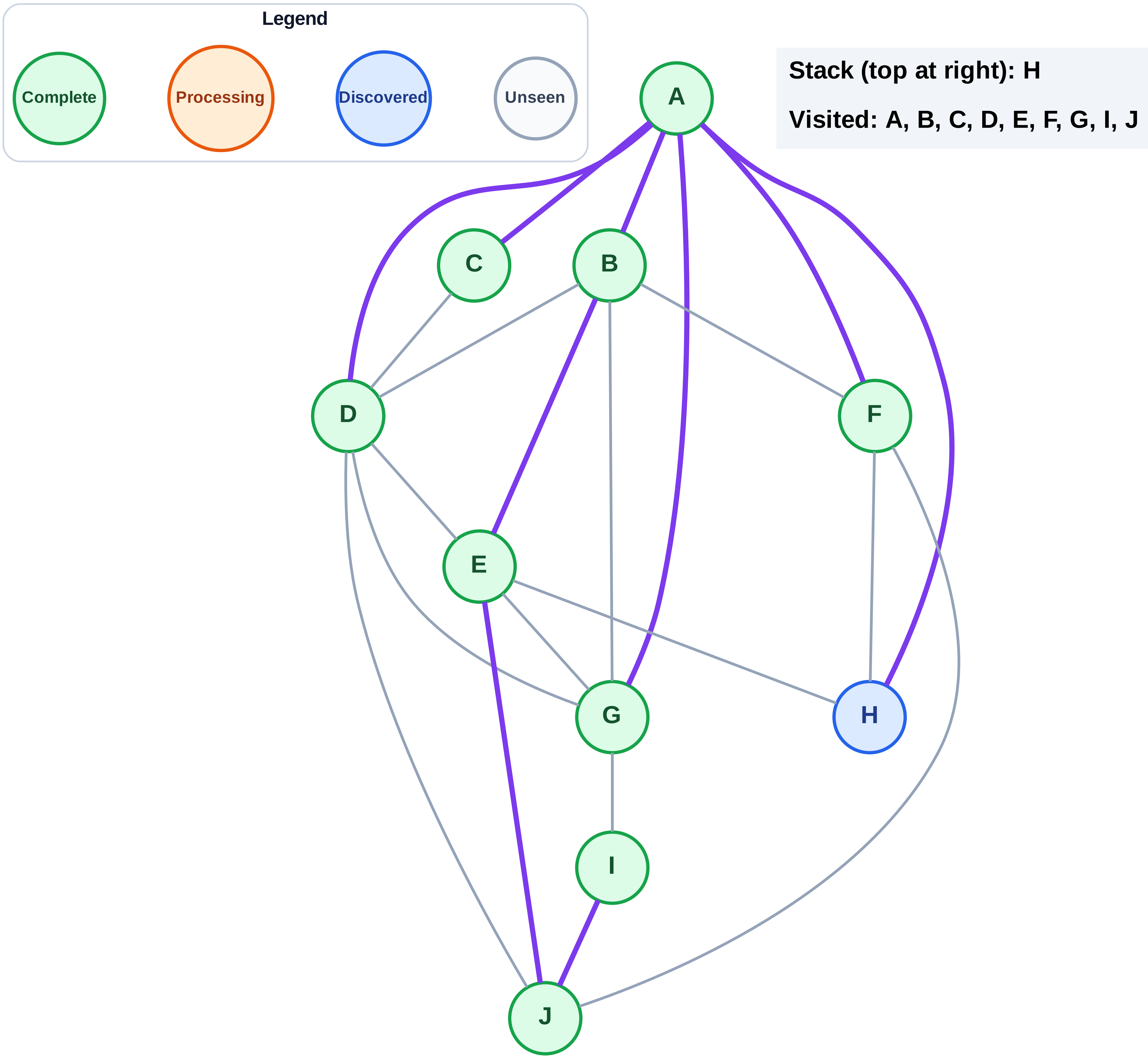



DFS Traversal
Step 17: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | G
Visited: A, B, C, D, E, F, I, J
A
B
C
D
E
F
G
H
I
J


DFS Traversal
Step 18: Process node G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J
A
B
C
D
E
F
G
H
I
J

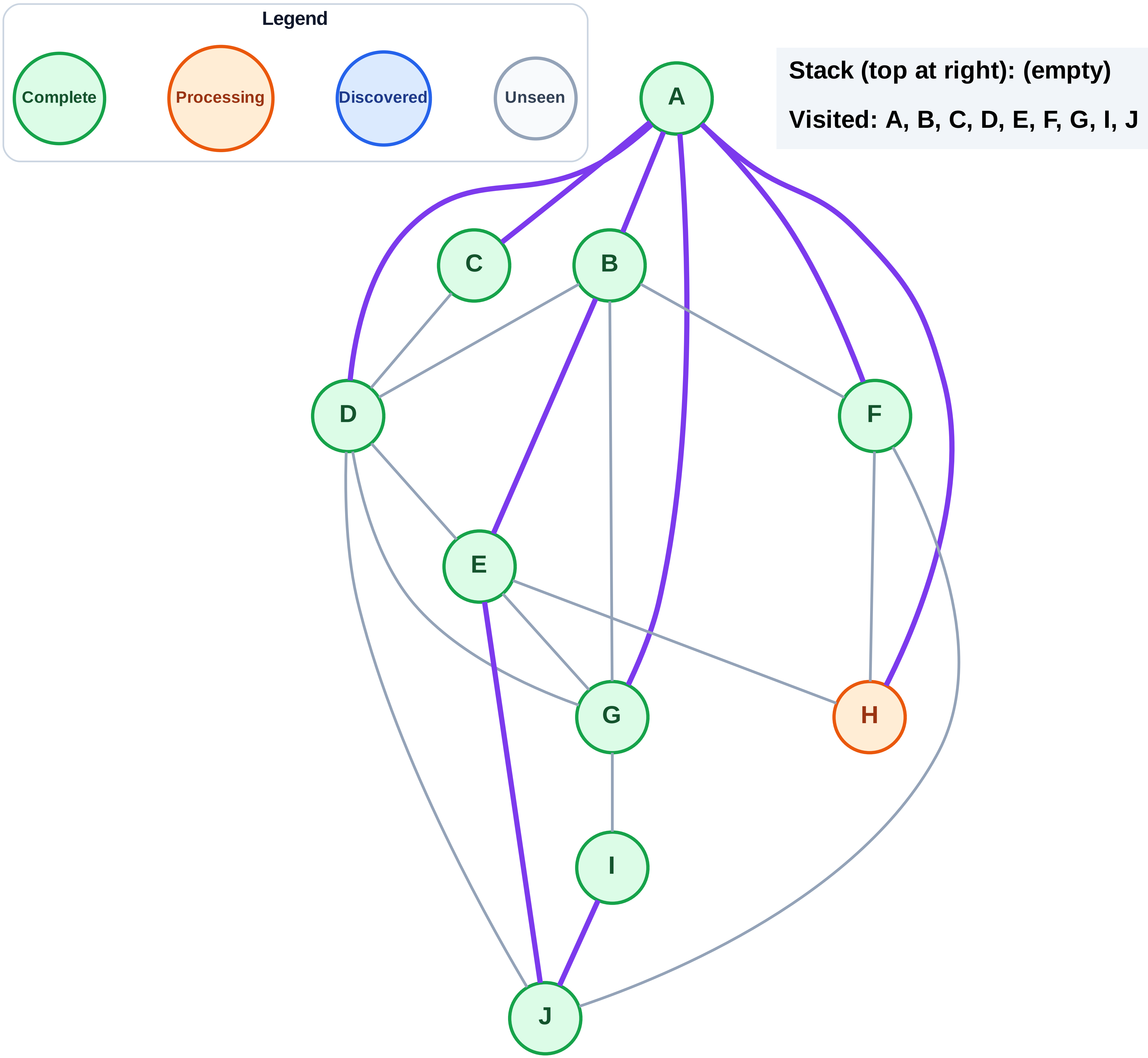
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H